codex-windows / 019f2174-3b37-7132-b85d-6b2dd1f1c4c9

Metadata

- Source: `codex-windows`
- Kind: `codex`
- Source file: `C:\Users\avidu\.codex\archived_sessions\rollout-2026-07-02T11-41-33-019f2174-3b37-7132-b85d-6b2dd1f1c4c9.jsonl`
- SHA-256: `560a0f3d4f7bb380997e11880939330d1b33af8d70e8c9246a02637250dfda05`
- Source modified: `2026-07-02T06:11:33+00:00`
- Imported at: `2026-07-05T16:48:16+00:00`
- cli_version: `0.142.5`
- cwd: `C:\Users\avidu\Projects\badminton-highlight-indexer`
- model_provider: `openai`
- session_id: `019f2174-3b37-7132-b85d-6b2dd1f1c4c9`
- source: `vscode`

Transcript

1. user (2026-07-02T06:11:33.305Z)

In the badminton-highlight-indexer repo, two golden-feature loaders have the SAME latent bug just fixed in `backend/eval/ablation.py::load_videos` (commit on branch `fix/ablation-litmus-mixed-schema`): they read `c["shuttle"]`/`c["person"]` unconditionally on every candidate, so they crash with `KeyError: 'shuttle'` on a MIXED golden corpus where some `output/*_golden_features.json` files carry the leaner velocity-windowing/distill schema (keys: active_frame_count, chunk_index, end, mean_velocity, peak_velocity, start, track_id_count ? no shuttle/person/audio cue dicts).

Affected:
- `backend/eval/fusion_compare.py::load_videos` (~line 39-48) ? `python -m backend.eval.fusion_compare --features-dir output` would KeyError on the n=11 corpus.
- `backend/eval/fusion_audio.py::load_videos_with_audio` (~line 56-69) ? same, plus it reads `c["audio"]` unconditionally too.

Unlike the ablation litmus, these are NOT caught by any test today because `tests/test_fusion_compare.py` and `tests/test_audio_features.py` only exercise synthetic tmp_path corpora, never the real `output/`. So this is latent, not currently red ? but it bites the moment someone runs those CLIs on the real mixed corpus.

Fix: mirror the ablation fix ? reuse/duplicate the `_is_fusion_schema(candidates)` predicate (full-fusion iff every candidate has `shuttle`+`person`; audio optional) and skip leaner windowing-only files with an info log instead of crashing. Consider whether `_is_fusion_schema` should move to a shared spot (e.g. it currently lives in `backend/eval/ablation.py`) vs. being duplicated ? keep it simple, DRY only if it doesn't create an awkward import. Add a regression test per loader (mixed corpus loads only the fusion files). Keep it offline/mocked.

Verify: full suite (`python run_all_tests.py`), `ruff check .`, `mypy backend training` all clean; coverage stays ?70 (currently ~80.7%). Branch from `origin/master` in an isolated worktree, re-verify HEAD before commit, stage explicit paths (shared checkout ? sibling sessions share it). ONE PR per work item; do not stack on the ablation fix.

2. assistant (2026-07-02T06:11:33.305Z)

I'll start by understanding the existing fix in `ablation.py` and the two affected loaders, then set up an isolated worktree off `origin/master`.

3. assistant (2026-07-02T06:11:33.305Z)

```
[external_agent_tool_call: Bash]
description: Fetch origin and show current state
command: git fetch origin --quiet && git log --oneline -1 origin/master && echo "---current branch---" && git branch --show-current && echo "---status---" && git status --short | head -20
```

[/external_agent_tool_call]

4. assistant (2026-07-02T06:11:33.305Z)

```
[external_agent_tool_result]
be39a70 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17) (#206)
---current branch---
docs/golden-regression-fixtures-plan
---status---
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
[/external_agent_tool_result]
```

5. assistant (2026-07-02T06:11:33.305Z)

Let me examine the existing fix in `ablation.py` and the two affected files.

6. assistant (2026-07-02T06:11:33.305Z)

```
[external_agent_tool_call: Bash]
description: Check ablation.py on origin/master for the fix
command: git show origin/master:backend/eval/ablation.py 2>/dev/null | grep -n
"_is_fusion_schema\|def load_videos\|shuttle\|person\|skip\|windowing" | head -40
[/external_agent_tool_call]
```

7. assistant (2026-07-02T06:11:33.305Z)

```
[external_agent_tool_result]
25:A signal can be a **feature column** (person/audio/shuttle columns the learned scorer
weights) or
35:    the golden corpus may lack (Gate-A vs held-shuttle FPs) is marked ``structural`` and is
NEVER
62:# Source: distill_local.FEATURE_NAMES (shuttle, includes net_crossings), fusion_features
63:# (person), audio_features (audio). The shuttle group's velocity block + duration are the
64:# learned-column shuttle signals; net_crossings is BOTH a column and the Gate-A gate knob.
68:                                      "in_court_ratio", "shuttle_player_colocation")
90:    - ``name`` ? table/row identity ("person.colocation" | "gate_a" | "audio").
91:    - ``group`` ? producer ("person" | "gate_a" | "audio" | "shuttle"). LOGO ablates a whole
group.
93:      pure gate signal like Gate-A whose column is shared with the shuttle group).
353:    """The current 4-group inventory (LOSO granularity): shuttle (velocity block +
duration),
354:    gate_a (the structural net-crossings gate AND its column), person, audio.
356:    Gate-A is marked ``structural`` (the held-shuttle FP guard) so it is reported but never
360:        Signal("shuttle.duration", "shuttle", ("duration",)),
361:        Signal("shuttle.velocity", "shuttle", ("peak_velocity", "mean_velocity",
"active_ratio")),
363:        Signal("person", "person", PERSON_COLUMNS),
386:def load_videos(features_dir: str) -> List[VideoCandidates]:
387:    """Load every ``*_golden_features.json`` into a ``VideoCandidates`` (intervals + shuttle
+
388:    person + audio columns + golden gts). Audio columns are optional (older JSONs predate
the
399:            features[fn] = [float(c["shuttle"][fn]) for c in cands]
401:            features[fn] = [float(c["person"][fn]) for c in cands]
411:    """The learned full-set reference: all 13 columns (shuttle 5 + person 5 + audio 3)
weighted
553:        "corpus may lack (e.g. Gate-A vs held-shuttle false positives); reported honestly
but "
[/external_agent_tool_result]
```

8. assistant (2026-07-02T06:11:33.305Z)

[external_agent_tool_call: Bash]

```
description: Find the ablation fix branch and _is_fusion_schema usage
command: git branch -a | grep -i ablation; echo "---"; grep -rn "_is_fusion_schema" backend/
2>/dev/null
[/external_agent_tool_call]
```

9. assistant (2026-07-02T06:11:33.305Z)

```
[external_agent_tool_result]
+ fix/ablation-litmus-mixed-schema
---
[/external_agent_tool_result]
```

10. assistant (2026-07-02T06:11:33.305Z)

`_is_fusion_schema` isn't on my current branch. Let me look at the fix branch to understand the pattern, and check whether it's merged to master.

11. assistant (2026-07-02T06:11:33.305Z)

```
[external_agent_tool_call: Bash]
description: Check if fix branch is merged and view its ablation diff
command: echo "=== is fix branch merged to origin/master? ==="; git branch --contains $(git rev-
parse fix/ablation-litmus-mixed-schema) -a 2>/dev/null | grep -i master || echo "NOT merged to
master"; echo; echo "=== diff of fix branch vs origin/master on ablation.py ==="; git diff
origin/master...fix/ablation-litmus-mixed-schema -- backend/eval/ablation.py
[/external_agent_tool_call]
```

12. assistant (2026-07-02T06:11:33.305Z)

```
[external_agent_tool_result]
=== is fix branch merged to origin/master? ===
NOT merged to master

=== diff of fix branch vs origin/master on ablation.py ===
diff --git a/backend/eval/ablation.py b/backend/eval/ablation.py
index fec3e7d..968eefd 100644
--- a/backend/eval/ablation.py
+++ b/backend/eval/ablation.py
@@ -383,16 +383,37 @@ def litmus_config() -> AblationConfig:
 # ---------------------------------------------------------------------------- corpus I/O


+def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) -> bool:
+    """True iff every candidate row carries the per-cue feature dicts the ablation scorer reads
+    (``shuttle`` + ``person``; ``audio`` stays optional ? older JSONs predate it).
+
+    The velocity-windowing / distill path persists a LEANER ``*_golden_features.json`` row
+    (``start``/``end`` + velocity stats, no cue dicts) for clips that never went through fusion
+    feature extraction. ``load_videos`` SKIPS those rather than crash on ``c["shuttle"]`` ? so
a
+    mixed corpus (some fusion files, some windowing-only) loads the ablatable subset cleanly.
+    An empty candidate list is not ablatable either."""
+    return bool(candidates) and all("shuttle" in c and "person" in c for c in candidates)
+
+
 def load_videos(features_dir: str) -> List[VideoCandidates]:
-    """Load every ``*_golden_features.json`` into a ``VideoCandidates`` (intervals + shuttle +
-    person + audio columns + golden gts). Audio columns are optional (older JSONs predate the
-    audio augment) ? a missing block defaults those columns to 0.0 so the loader never crashes,
-    mirroring ``experiment._feature_column``'s tolerance."""
+    """Load every FULL-FUSION-SCHEMA ``*_golden_features.json`` into a ``VideoCandidates``
+    (intervals + shuttle + person + audio columns + golden gts). Audio columns are optional
+    (older JSONs predate the audio augment) ? a missing block defaults those columns to 0.0 so
+    the loader never crashes, mirroring ``experiment._feature_column``'s tolerance.
```

```
+
+    Leaner velocity-windowing-only files (no ``shuttle``/``person`` cue dicts ? the schema the
+    R3/R4 corpus-growth path persists) are SKIPPED, not crashed on, so a mixed corpus loads the
+    ablatable subset cleanly (see ``_is_fusion_schema``)."""
     videos: List[VideoCandidates] = []
     for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
         with open(path, encoding="utf-8") as f:
             d = json.load(f)
         cands = d["candidates"]
+        if not _is_fusion_schema(cands):
+            logger.info("ablation.load_videos: skipping %s ? windowing-only schema "
+                        "(no shuttle/person cue dicts), not an ablatable fusion-feature file",
+                        os.path.basename(path))
+            continue
         intervals = [(float(c["start"]), float(c["end"])) for c in cands]
         features: Dict[str, List[float]] = {}
         for fn in SHUTTLE_COLUMNS:
[/external_agent_tool_result]
```

13. assistant (2026-07-02T06:11:33.305Z)

The fix branch is **not** merged to master, so importing `_is_fusion_schema` from `ablation.py` would stack on an unmerged branch ? I'll keep it self-contained. Let me read both affected files in full.

14. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\fusion_compare.py
[/external_agent_tool_call]
```

15. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_result]
"""Pure-CPU LOVO comparison: shuttle-only vs shuttle+person learned fusion on the golden set.

The fusion-P3 measurement. Loads the per-video feature JSONs written by `fusion_golden.py`
and runs the SHIPPED experiment harness (`experiment.compare`, honest leave-one-video-out) ?
no new scoring/labeling math. The two variants differ ONLY by `feature_names` (the harness's
designed seam): the person features are extra columns the learned LogisticRegression weights,
so the held-out ?F1 is an honest answer to "does the person cue help?" ? not a hunch.

Run (no GPU):
    python -m backend.eval.fusion_compare --features-dir output --record
"""
from __future__ import annotations

import argparse
import glob
import json
import os
from typing import Any, Dict, List

from backend.eval import experiment
from backend.eval.eval_stats import paired_delta_ci
from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES


def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
```

```
    """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign test
    (eval_stats, C1 ? "never a bare mean"). Aligns the two variants' per-video held-out scores
    by video name, so the headline ?F1 comes with how-wide + did-it-win-on-enough-videos."""
    by_a = {p["video"]: p[metric] for p in res["variant_a"]["per_video"]}
    by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
    vids = [p["video"] for p in res["variant_a"]["per_video"]]
    return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])


def load_videos(features_dir: str) -> List[experiment.VideoCandidates]:
    """Load every ``*_golden_features.json`` into a VideoCandidates (intervals + the 10
    feature columns + golden gts). The harness derives labels from gts itself."""
    videos: List[experiment.VideoCandidates] = []
    for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
        with open(path, encoding="utf-8") as f:
            d = json.load(f)
        cands = d["candidates"]
        intervals = [(float(c["start"]), float(c["end"])) for c in cands]
        features: Dict[str, List[float]] = {}
        for fn in SHUTTLE_FEATURE_NAMES:
            features[fn] = [float(c["shuttle"][fn]) for c in cands]
        for fn in PERSON_FEATURE_NAMES:
            features[fn] = [float(c["person"][fn]) for c in cands]
        gts = [(float(a), float(b)) for a, b in d["gts"]]
        videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
                                                 features=features, gts=gts))
    return videos


def compare(videos: List[experiment.VideoCandidates], iou: float = 0.5,
            threshold: float = 0.5) -> Dict[str, Any]:
    shuttle_only = experiment.Variant(name="shuttle_only", feature_names=list(SHUTTLE_FEATURE_NAMES),
                                      threshold=threshold)
    fusion = experiment.Variant(name="shuttle_person_fusion",
                                feature_names=list(SHUTTLE_FEATURE_NAMES) + list(PERSON_FEATURE_NAMES),
                                threshold=threshold)
    return experiment.compare(videos, shuttle_only, fusion, iou=iou, honest=True)


def format_result(res: Dict[str, Any], iou: float) -> str:
    a, b, d = res["variant_a"]["aggregate"], res["variant_b"]["aggregate"], res["delta"]
    ds = delta_stats(res)
    st = ds["sign_test"]
    lines = [
        f"=== Fusion P3 LOVO (n={res['num_videos']} golden videos, IoU>={iou}, honest) ===",
        f"  shuttle-only : F1={a['f1']:.3f}  P={a['precision']:.3f}  R={a['recall']:.3f}  IoU={a['mean_iou']:.3f}",
        f"  +person      : F1={b['f1']:.3f}  P={b['precision']:.3f}  R={b['recall']:.3f}  IoU={b['mean_iou']:.3f}",
      ...
```
[/external_agent_tool_result]

16. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\fusion_audio.py
[/external_agent_tool_call]
```

17. assistant (2026-07-02T06:11:33.306Z)

[external_agent_tool_result]

```
"""Fusion P4 ? measure the INCREMENTAL audio-cue lift on top of the person cue (CPU-only).

Reuses the person-feature JSONs from `fusion_golden.py` (no GPU re-detect): `augment` runs
`detect_hits_in_video` once per video (ffmpeg + numpy, CPU) and adds the 3 audio features per
window; `compare_increment` then runs the SHIPPED harness between a shuttle+person variant
and a shuttle+person+audio variant ? so audio is measured as a marginal contribution ON TOP
of person, not against raw shuttle. Audio enters only as columns the learned scorer weights
(no special-casing, default-OFF); the golden-set ?F1 decides whether it earns its keep.

Run (no GPU):
    python -m backend.eval.fusion_audio --manifest output/human_lovo_manifest.json --features-dir output --augment --compare
"""
from __future__ import annotations

import argparse
import glob
import json
import os
from typing import Any, Dict, List

from backend.eval import experiment
from backend.eval.fusion_compare import delta_stats
from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES, _derive_video_path
from backend.pipeline.detectors import audio_features as af
from backend.pipeline.audio.hit_detector import detect_hits_in_video


def augment(manifest_path: str, features_dir: str, k: float = 2.5) -> List[str]:
    """Add the 3 audio features to each existing ``<name>_golden_features.json`` window."""
    with open(manifest_path, encoding="utf-8") as f:
        specs = json.load(f)
    written: List[str] = []
    for spec in specs:
        jp = os.path.join(features_dir, f"{spec['name']}_golden_features.json")
        if not os.path.isfile(jp):
            print(f"[fusion_audio] {spec['name']}: no feature JSON ? run fusion_golden first; skipping")
            continue
        video = _derive_video_path(spec)
        print(f"[fusion_audio] {spec['name']}: detecting audio hits (k={k})...", flush=True)
        hits = detect_hits_in_video(video, k=k)
        with open(jp, encoding="utf-8") as f:
            data = json.load(f)
        for c in data["candidates"]:
            c["audio"] = af.compute_audio_features(hits, c["start"], c["end"])
        with open(jp, "w", encoding="utf-8") as f:
            json.dump(data, f, indent=2)
        print(f"[fusion_audio] {spec['name']}: {len(hits)} hits over {len(data['candidates'])} windows", flush=True)
        written.append(jp)
    return written


def load_videos_with_audio(features_dir: str) -> List[experiment.VideoCandidates]:
```

```
    """Load the feature JSONs into VideoCandidates including the audio columns (requires
    `augment` to have run ? raises if a JSON lacks audio)."""
    videos: List[experiment.VideoCandidates] = []
    for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
        with open(path, encoding="utf-8") as f:
            d = json.load(f)
        cands = d["candidates"]
        if cands and "audio" not in cands[0]:
            raise SystemExit(f"{path} has no audio features ? run `--augment` first")
        intervals = [(float(c["start"]), float(c["end"])) for c in cands]
        features: Dict[str, List[float]] = {}
        for fn in SHUTTLE_FEATURE_NAMES:
            features[fn] = [float(c["shuttle"][fn]) for c in cands]
        for fn in PERSON_FEATURE_NAMES:
            features[fn] = [float(c["person"][fn]) for c in cands]
        for fn in af.AUDIO_FEATURE_NAMES:
            features[fn] = [float(c["audio"][fn]) for c in cands]
        gts = [(float(a), float(b)) for a, b in d["gts"]]
        videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
                                                 features=features, gts=gts))
    return videos


def compare_increment(videos: List[experiment.VideoCandidates], iou: f...
```
[/external_agent_tool_result]

18. assistant (2026-07-02T06:11:33.306Z)

Both files use `from typing import Any, Dict, List`, and `fusion_audio` already imports `delta_stats` from `fusion_compare` ? so importing a shared predicate from `fusion_compare` is a natural, non-awkward dependency. Let me check the logger setup in `ablation.py` and the existing tests.

19. assistant (2026-07-02T06:11:33.306Z)

[external_agent_tool_call: Bash]
description: Check logger setup in ablation and fusion_compare
command: grep -n "import logging\|^logger\|logger =" backend/eval/ablation.py | head; echo "=== fusion_compare logger? ==="; grep -n "import logging\|logger" backend/eval/fusion_compare.py | head; echo "=== Sequence import in ablation ==="; grep -n "from typing import" backend/eval/ablation.py
[/external_agent_tool_call]

20. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_result]
51:import logging
59:logger = logging.getLogger(__name__)
=== fusion_compare logger? ===
=== Sequence import in ablation ===
54:from typing import Any, Dict, List, Optional, Sequence, Tuple
[/external_agent_tool_result]
```

21. assistant (2026-07-02T06:11:33.306Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_fusion_compare.py
[/external_agent_tool_call]

22. assistant (2026-07-02T06:11:33.306Z)

[external_agent_tool_result]

```
"""GPU-free tests for the fusion-P3 measurement (fusion_golden extractor +
fusion_compare LOVO).
```

```python

The LOVO driver is exercised on a synthetic 6-video set where the SHUTTLE features are
non-separating (constant) but a PERSON feature cleanly separates rallies ? so the harness must
report a positive fusion ?F1. The extractor is exercised with a stubbed detector (no GPU).
"""
import json
import os
from types import SimpleNamespace

import pytest

from backend.eval import fusion_compare
from backend.eval.fusion_compare import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES


def _write_video(d, name):
    """A video whose shuttle features are identical for pos/neg (no signal) but whose
    players_in_court separates rallies (4) from junk (0)."""
    cands = []
    for (s, e), pos in [((0.0, 10.0), 1), ((20.0, 30.0), 1), ((40.0, 41.0), 0), ((50.0, 51.0), 0)]:
        shuttle = {fn: 1.0 for fn in SHUTTLE_FEATURE_NAMES}                 # constant -> non-separating
        person = {fn: 0.0 for fn in PERSON_FEATURE_NAMES}
        person["players_in_court"] = 4.0 if pos else 0.0                   # the separating signal
        person["in_court_ratio"] = 1.0 if pos else 0.0
        cands.append({"start": s, "end": e, "shuttle": shuttle, "person": person, "label": pos})
    data = {"name": name, "fps": 30.0, "frame_width": 1920.0,
            "candidates": cands, "gts": [[0.0, 10.0], [20.0, 30.0]]}
    with open(os.path.join(d, f"{name}_golden_features.json"), "w", encoding="utf-8") as f:
        json.dump(data, f)


def test_fusion_compare_reports_person_driven_lift(tmp_path):
    d = str(tmp_path)
    for k in range(6):
        _write_video(d, f"vid{k}")
    videos = fusion_compare.load_videos(d)
    assert len(videos) == 6
    res = fusion_compare.compare(videos, iou=0.5)
    assert res["num_videos"] == 6
    assert res["variant_b"]["honest_lovo"] is True                 # LOVO-honest learned variant
    # Fusion (with the separating person feature) must beat shuttle-only (no signal).
    assert res["delta"]["f1"] > 0.0
    assert res["variant_b"]["aggregate"]["f1"] > res["variant_a"]["aggregate"]["f1"]
    # format_result must not crash and must mention both variants.
    out = fusion_compare.format_result(res, 0.5)
    assert "shuttle-only" in out and "+person" in out


def test_fusion_compare_delta_stats_and_ci(tmp_path):
    """delta_stats yields a paired bootstrap CI + sign test; format_result reports them."""
    d = str(tmp_path)
    for k in range(6):
        _write_video(d, f"vid{k}")
    res = fusion_compare.compare(fusion_compare.load_videos(d), iou=0.5)
    ds = fusion_compare.delta_stats(res)
    assert set(ds) >= {"mean_delta", "ci_low", "ci_high", "sign_test",
```

```
"ci_excludes_zero"}
        assert ds["ci_low"] - 1e-9 <= ds["mean_delta"] <= ds["ci_high"] + 1e-9   # CI brackets mean (±fp)
        assert ds["sign_test"]["wins"] == 6          # fusion wins on every video (person feature separates)
        assert ds["ci_excludes_zero"] is True
        out = fusion_compare.format_result(res, 0.5)
        assert "CI" in out and "sign test" in out


    def test_fusion_compare_needs_two_videos(tmp_path):
        _write_video(str(tmp_path), "only")
        videos = fusion_compare.load_videos(str(tmp_path))
        assert len(videos) == 1  # compare() / experiment.compare needs >=2 for honest LOVO


    # ----------------------------------------------------------------- extractor assembly

    def test_extract_video_assembles_shuttle_and_person(monkeypatch):
        from backend.eval import fusion_golden
        from backend.pipeline.detectors.tracknet_runner import TrackNetRunner

        intervals = [(0.0, 10.0), (40.0, 41.0)]
        x_shuttle = [[10.0, 0.3, 0.2, 0.5, 4.0], [1.0, 0.1, 0.05, 0.2, 0.0]]
        points = [SimpleNamespace(frame=i, visible=True, x=500.0, y=(...
[/external_agent_tool_result]
```

23. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_audio_features.py
[/external_agent_tool_call]
```

24. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_result]
    """Tests for the P4 audio cue ? pure audio features + the incremental-lift driver (no GPU/ffmpeg)."""
    import json
    import os
    from types import SimpleNamespace

    import pytest

    from backend.eval import fusion_audio
    from backend.pipeline.detectors import audio_features as af


    def H(t, strength=1.0):
        return SimpleNamespace(t=t, strength=strength)


    def test_compute_audio_features_empty():
        f = af.compute_audio_features([], 0.0, 10.0)
        assert set(f) == set(af.AUDIO_FEATURE_NAMES)
        assert all(v == 0.0 for v in f.values())


    def test_compute_audio_features_counts_rate_strength():
        hits = [H(1.0, 2.0), H(2.0, 4.0), H(5.0, 6.0), H(50.0, 1.0)]   # 3 in [0,10), 1 outside
        f = af.compute_audio_features(hits, 0.0, 10.0)
        assert f["audio_hit_count"] == 3.0
        assert f["audio_hit_rate"] == pytest.approx(0.3)              # 3 hits / 10 s
        assert f["audio_hit_strength_mean"] == pytest.approx((2 + 4 + 6) / 3)
```

```python


def test_compute_audio_features_half_open_boundary():
    hits = [H(10.0), H(20.0)]                                     # window [10,20): 10 in, 20 out
    f = af.compute_audio_features(hits, 10.0, 20.0)
    assert f["audio_hit_count"] == 1.0


# ----------------------------------------------------- incremental-lift driver (synthetic)

def _write_video(d, name):
    """shuttle + person constant (no signal); audio_hit_count separates rallies (8) from junk (0)."""
    cands = []
    for (s, e), pos in [((0.0, 10.0), 1), ((20.0, 30.0), 1), ((40.0, 41.0), 0), ((50.0, 51.0), 0)]:
        shuttle = {fn: 1.0 for fn in fusion_audio.SHUTTLE_FEATURE_NAMES}
        person = {fn: 1.0 for fn in fusion_audio.PERSON_FEATURE_NAMES}
        audio = {fn: 0.0 for fn in af.AUDIO_FEATURE_NAMES}
        audio["audio_hit_count"] = 8.0 if pos else 0.0
        cands.append({"start": s, "end": e, "shuttle": shuttle, "person": person,
                      "audio": audio, "label": pos})
    data = {"name": name, "fps": 30.0, "frame_width": 1920.0,
            "candidates": cands, "gts": [[0.0, 10.0], [20.0, 30.0]]}
    with open(os.path.join(d, f"{name}_golden_features.json"), "w", encoding="utf-8") as f:
        json.dump(data, f)


def test_audio_increment_compare(tmp_path):
    d = str(tmp_path)
    for k in range(6):
        _write_video(d, f"v{k}")
    videos = fusion_audio.load_videos_with_audio(d)
    assert len(videos) == 6
    res = fusion_audio.compare_increment(videos, iou=0.5)
    assert res["variant_b"]["honest_lovo"] is True
    assert res["delta"]["f1"] > 0.0          # audio adds separating signal on top of constant shuttle+person
    assert "audio" in fusion_audio.format_result(res, 0.5).lower()


def test_load_videos_requires_audio(tmp_path):
    data = {"name": "x", "fps": 30.0, "frame_width": 1920.0,
            "candidates": [{"start": 0.0, "end": 1.0,
                            "shuttle": {fn: 0.0 for fn in fusion_audio.SHUTTLE_FEATURE_NAMES},
                            "person": {fn: 0.0 for fn in fusion_audio.PERSON_FEATURE_NAMES},
                            "label": 0}],
            "gts": [[0.0, 1.0]]}
    with open(os.path.join(str(tmp_path), "x_golden_features.json"), "w", encoding="utf-8") as f:
        json.dump(data, f)
    with pytest.raises(SystemExit):
        fusion_audio.load_videos_with_audio(str(tmp_path))

```
[/external_agent_tool_result]

25. assistant (2026-07-02T06:11:33.306Z)

Now I understand the full picture. Plan:
- Add a module-level `logger` + a self-contained `_is_fusion_schema` predicate to

`fusion_compare.py` (its natural home ? `fusion_audio` already imports from it). I'll **duplicate** rather than import from `ablation.py` because that fix is unmerged on master; importing would stack on it.
- `fusion_compare.load_videos`: skip non-fusion files with an info log.
- `fusion_audio.load_videos_with_audio`: import the predicate, skip windowing-only files **before** the audio-missing check (so genuine fusion-without-audio still raises).
- Add a regression test per loader.

Let me set up an isolated worktree off `origin/master`.

26. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_call: Bash]
description: Create isolated worktree off origin/master
command: git worktree add -b fix/fusion-loaders-mixed-schema "../bhi-fusion-loaders"
origin/master 2>&1 | tail -5 && echo "=== worktree HEAD ===" && git -C "../bhi-fusion-loaders"
log --oneline -1 && git -C "../bhi-fusion-loaders" branch --show-current
[/external_agent_tool_call]
```

27. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_result]
Preparing worktree (new branch 'fix/fusion-loaders-mixed-schema')
branch 'fix/fusion-loaders-mixed-schema' set up to track 'origin/master'.
HEAD is now at be39a70 docs: rally-detection quality report ? technical-paper checkpoint
(2026-06-17) (#206)
=== worktree HEAD ===
be39a70 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17) (#206)
fix/fusion-loaders-mixed-schema
[/external_agent_tool_result]
```

28. assistant (2026-07-02T06:11:33.306Z)

```
[external_agent_tool_call: Bash]
description: Confirm worktree files are at master version
command: WT="../bhi-fusion-loaders"; echo "=== fusion_compare diff (worktree vs what I read)
==="; diff <(git show be39a70:backend/eval/fusion_compare.py)
"$WT/backend/eval/fusion_compare.py" && echo "IDENTICAL to master"; echo "=== confirm
_is_fusion_schema absent in worktree ==="; grep -rn "_is_fusion_schema" "$WT/backend/eval/" ||
echo "absent (as expected ? ablation fix unmerged)"
[/external_agent_tool_call]
```

29. assistant (2026-07-02T06:11:33.307Z)

```
[external_agent_tool_result]
=== fusion_compare diff (worktree vs what I read) ===
1,102c1,102
< """Pure-CPU LOVO comparison: shuttle-only vs shuttle+person learned fusion on the golden set.
<
< The fusion-P3 measurement. Loads the per-video feature JSONs written by `fusion_golden.py`
< and runs the SHIPPED experiment harness (`experiment.compare`, honest leave-one-video-out) ?
< no new scoring/labeling math. The two variants differ ONLY by `feature_names` (the harness's
< designed seam): the person features are extra columns the learned LogisticRegression weights,
< so the held-out ?F1 is an honest answer to "does the person cue help?" ? not a hunch.
<
< Run (no GPU):
<     python -m backend.eval.fusion_compare --features-dir output --record
< """
< from __future__ import annotations
<
< import argparse
< import glob
< import json
< import os
```

```
< from typing import Any, Dict, List
<
< from backend.eval import experiment
< from backend.eval.eval_stats import paired_delta_ci
< from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES
<
<
< def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
<     """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign test
<     (eval_stats, C1 ? "never a bare mean"). Aligns the two variants' per-video held-out scores
<     by video name, so the headline ?F1 comes with how-wide + did-it-win-on-enough-videos."""
<     by_a = {p["video"]: p[metric] for p in res["variant_a"]["per_video"]}
<     by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
<     vids = [p["video"] for p in res["variant_a"]["per_video"]]
<     return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])
<
<
< def load_videos(features_dir: str) -> List[experiment.VideoCandidates]:
<     """Load every ``*_golden_features.json`` into a VideoCandidates (intervals + the 10
<     feature columns + golden gts). The harness derives labels from gts itself."""
<     videos: List[experiment.VideoCandidates] = []
<     for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
<         with open(path, encoding="utf-8") as f:
<             d = json.load(f)
<         cands = d["candidates"]
<         intervals = [(float(c["start"]), float(c["end"])) for c in cands]
<         features: Dict[str, List[float]] = {}
<         for fn in SHUTTLE_FEATURE_NAMES:
<             features[fn] = [float(c["shuttle"][fn]) for c in cands]
<         for fn in PERSON_FEATURE_NAMES:
<             features[fn] = [float(c["person"][fn]) for c in cands]
<         gts = [(float(a), float(b)) for a, b in d["gts"]]
<         videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
<                                                  features=features, gts=gts))
<     return videos
<
<
< def compare(videos: List[experiment.VideoCandidates], iou: float = 0.5,
<             threshold: float = 0.5) -> Dict[str, Any]:
<     shuttle_only = experiment.Variant(name="shuttle_only",
feature_names=list(SHUTTLE_FEATURE_NAMES),
<                                       threshold=threshold)
<     fusion = experiment.Variant(name="shuttle_person_fusion",
<                                 feature_names=list(SHUTTLE_FEATURE_NAMES) +
list(PERSON_FEATURE_NAMES),
<                                 threshold=threshold)
<     return experiment.compare(videos, shuttle_only, fusion, iou=iou, honest=True)
<
<
< def format_result(res: Dict[str, Any], iou: float) -> str:
<     a, b, d = res["variant_a"]["aggregate"], res["variant_b"]["aggregate"], res["delta"]
<     ds = delta_stats(res)
<     st = ds["sign_test"]
<     lines = [
<         f"=== Fusion P3 LOVO (n={res['num_videos']} golden videos, IoU>={iou}, honest) ===",
<         f"  shuttle-only : F1={a['f1']:.3f}  P={a['precision']:.3f}  R={a['recall']:.3f}
IoU={a['mean_iou']:.3f}",
<         f"  +person      : F1={b['f1']:.3f}  P={b['precision']:.3f}  R={b['recall']:.3f}
IoU={b['mean_iou']:.3f}",
<     ...
[/external_agent_tool_result]
```

30. assistant (2026-07-02T06:11:33.307Z)

Content is identical (the diff is just CRLF vs LF display). Let me check line-ending config so my edits don't introduce noise.

31. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_call: Bash]
description: Check line-ending configuration
command: cd "../bhi-fusion-loaders" && echo "=== .gitattributes ==="; cat .gitattributes
2>/dev/null || echo "(none)"; echo "=== autocrlf ==="; git config --get core.autocrlf; echo "===
file CRLF check ==="; file backend/eval/fusion_compare.py 2>/dev/null; git ls-files --eol
backend/eval/fusion_compare.py backend/eval/fusion_audio.py
[/external_agent_tool_call]

32. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_result]
=== .gitattributes ===

Golden Regression Fixtures (M1) ? pin LF line endings so the committed fixtures are byte-stable across OSes despite core.autocrlf. The characterization gate parse-compa (never byte-compares) JSON, but a stable on-disk form keeps `git diff` reviewable and fixtures reproducible. See docs/GOLDEN_REGRESSION_FIXTURES.md §4d/§4e.

eval_baselines/fixtures/** text eol=lf
=== autocrlf ===
true
=== file CRLF check ===
backend/eval/fusion_compare.py: Python script, Unicode text, UTF-8 text executable, with CRLF
line terminators
i/lf   w/crlf attr/   backend/eval/fusion_audio.py
i/lf   w/crlf attr/   backend/eval/fusion_compare.py
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

33. assistant (2026-07-02T06:11:33.307Z)

`i/lf w/crlf` with `autocrlf=true` ? git stores LF, so commits stay clean. Now let me read the worktree copies (required before editing them) and make the changes.

34. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_compare.py
[/external_agent_tool_call]

35. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_result]
1   """Pure-CPU LOVO comparison: shuttle-only vs shuttle+person learned fusion on the golden
set.
2
3   The fusion-P3 measurement. Loads the per-video feature JSONs written by
`fusion_golden.py`
4   and runs the SHIPPED experiment harness (`experiment.compare`, honest leave-one-video-
out) ?
5   no new scoring/labeling math. The two variants differ ONLY by `feature_names` (the
harness's
6   designed seam): the person features are extra columns the learned LogisticRegression
weights,
7   so the held-out ?F1 is an honest answer to "does the person cue help?" ? not a hunch.
8
9   Run (no GPU):
10   python -m backend.eval.fusion_compare --features-dir output --record
11   """

```
from __future__ import annotations

import argparse
import glob
import json
import os
from typing import Any, Dict, List

from backend.eval import experiment
from backend.eval.eval_stats import paired_delta_ci
from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES


def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
    """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign test
    (eval_stats, C1 ? "never a bare mean"). Aligns the two variants' per-video held-out scores
    by video name, so the headline ?F1 comes with how-wide + did-it-win-on-enough-videos."""
    by_a = {p["video"]: p[metric] for p in res["variant_a"]["per_video"]}
    by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
    vids = [p["video"] for p in res["variant_a"]["per_video"]]
    return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])


def load_videos(features_dir: str) -> List[experiment.VideoCandidates]:
    """Load every ``*_golden_features.json`` into a VideoCandidates (intervals + the 10
    feature columns + golden gts). The harness derives labels from gts itself."""
    videos: List[experiment.VideoCandidates] = []
    for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
        with open(path, encoding="utf-8") as f:
            d = json.load(f)
        cands = d["candidates"]
        intervals = [(float(c["start"]), float(c["end"])) for c in cands]
        features: Dict[str, List[float]] = {}
        for fn in SHUTTLE_FEATURE_NAMES:
            features[fn] = [float(c["shuttle"][fn]) for c in cands]
        for fn in PERSON_FEATURE_NAMES:
            features[fn] = [float(c["person"][fn]) for c in cands]
        gts = [(float(a), float(b)) for a, b in d["gts"]]
        videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
                                                 features=features, gts=gts))
    return videos
```
[/external_agent_tool_result]

36. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_compare.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

37. assistant (2026-07-02T06:11:33.307Z)

Now the edits to `fusion_compare.py`. First, imports + logger:

38. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_compare.py
[/external_agent_tool_call]

39. assistant (2026-07-02T06:11:33.307Z)

Now add the predicate and update `load_videos`:

40. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_compare.py
[/external_agent_tool_call]

41. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_compare.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

42. assistant (2026-07-02T06:11:33.307Z)

Now `fusion_audio.py`. Read the worktree copy's header, then edit:

43. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_audio.py
[/external_agent_tool_call]

44. assistant (2026-07-02T06:11:33.307Z)

[external_agent_tool_result]

```
"""Fusion P4 ? measure the INCREMENTAL audio-cue lift on top of the person cue (CPU-only).

Reuses the person-feature JSONs from `fusion_golden.py` (no GPU re-detect): `augment` runs
`detect_hits_in_video` once per video (ffmpeg + numpy, CPU) and adds the 3 audio features per
window; `compare_increment` then runs the SHIPPED harness between a shuttle+person variant
and a shuttle+person+audio variant ? so audio is measured as a marginal contribution ON TOP
of person, not against raw shuttle. Audio enters only as columns the learned scorer weights
(no special-casing, default-OFF); the golden-set ?F1 decides whether it earns its keep.

Run (no GPU):
    python -m backend.eval.fusion_audio --manifest output/human_lovo_manifest.json --features-dir output --augment --compare
"""
from __future__ import annotations

import argparse
import glob
import json
import os
from typing import Any, Dict, List

from backend.eval import experiment
from backend.eval.fusion_compare import delta_stats
from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES, _derive_video_path
from backend.pipeline.detectors import audio_features as af
from backend.pipeline.audio.hit_detector import detect_hits_in_video

```
[/external_agent_tool_result]

45. assistant (2026-07-02T06:11:33.307Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_audio.py
[/external_agent_tool_call]
```

46. assistant (2026-07-02T06:11:33.307Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_audio.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

47. assistant (2026-07-02T06:11:33.307Z)

Now add a logger to `fusion_audio` and update `load_videos_with_audio`. First the import/logger:

48. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_audio.py
[/external_agent_tool_call]
```

49. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_audio.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

50. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_audio.py
[/external_agent_tool_call]
```

51. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-fusion-loaders\backend\eval\fusion_audio.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

52. assistant (2026-07-02T06:11:33.308Z)

Now the regression tests. Let me read the worktree test files and add a test to each.

53. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\tests\test_fusion_compare.py
[/external_agent_tool_call]
```

54. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_result]
1       """GPU-free tests for the fusion-P3 measurement (fusion_golden extractor +
fusion_compare LOVO).
2
3       The LOVO driver is exercised on a synthetic 6-video set where the SHUTTLE features are
4       non-separating (constant) but a PERSON feature cleanly separates rallies ? so the
harness must
5       report a positive fusion ?F1. The extractor is exercised with a stubbed detector (no
```

```
GPU).
"""
import json
import os
from types import SimpleNamespace

import pytest

from backend.eval import fusion_compare
from backend.eval.fusion_compare import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES


def _write_video(d, name):
    """A video whose shuttle features are identical for pos/neg (no signal) but whose
    players_in_court separates rallies (4) from junk (0)."""
    cands = []
    for (s, e), pos in [((0.0, 10.0), 1), ((20.0, 30.0), 1), ((40.0, 41.0), 0), ((50.0, 51.0), 0)]:
        shuttle = {fn: 1.0 for fn in SHUTTLE_FEATURE_NAMES}                 # constant -> non-separating
        person = {fn: 0.0 for fn in PERSON_FEATURE_NAMES}
        person["players_in_court"] = 4.0 if pos else 0.0                   # the separating signal
        person["in_court_ratio"] = 1.0 if pos else 0.0
        cands.append({"start": s, "end": e, "shuttle": shuttle, "person": person, "label": pos})
    data = {"name": name, "fps": 30.0, "frame_width": 1920.0,
            "candidates": cands, "gts": [[0.0, 10.0], [20.0, 30.0]]}
    with open(os.path.join(d, f"{name}_golden_features.json"), "w", encoding="utf-8") as f:
        json.dump(data, f)


def test_fusion_compare_reports_person_driven_lift(tmp_path):
    d = str(tmp_path)
    for k in range(6):
        _write_video(d, f"vid{k}")
    videos = fusion_compare.load_videos(d)
    assert len(videos) == 6
    res = fusion_compare.compare(videos, iou=0.5)
    assert res["num_videos"] == 6
    assert res["variant_b"]["honest_lovo"] is True                 # LOVO-honest learned variant
    # Fusion (with the separating person feature) must beat shuttle-only (no signal).
    assert res["delta"]["f1"] > 0.0
    assert res["variant_b"]["aggregate"]["f1"] > res["variant_a"]["aggregate"]["f1"]
    # format_result must not crash and must mention both variants.
    out = fusion_compare.format_result(res, 0.5)
    assert "shuttle-only" in out and "+person" in out


def test_fusion_compare_delta_stats_and_ci(tmp_path):
    """delta_stats yields a paired bootstrap CI + sign test; format_result reports them."""
    d = str(tmp_path)
    for k in range(6):
        _write_video(d, f"vid{k}")
    res = fusion_compare.compare(fusion_compare.load_videos(d), iou=0.5)
    ds = fusion_compare.delta_stats(res)
    assert set(ds) >= {"mean_delta", "ci_low", "ci_high", "sign_test", "ci_excludes_zero"}
    assert ds["ci_low"] - 1e-9 <= ds["mean_delta"] <= ds["ci_high"] + 1e-9   # CI brackets mean (±fp)
    assert ds["sign_test"]["wins"] == 6          # fusion wins on every video (person feature separates)
```

```
60          assert ds["ci_excludes_zero"] is True
61          out = fusion_compare.format_result(res, 0.5)
62          assert "CI" in out and "sign test" in out
63
64
65      def test_fusion_compare_needs_two_videos(tmp_path):
66          _write_video(str(tmp_path), "only")
67          videos = fusion_compare.load_videos(str(tmp_path))
68          assert len(videos) == 1  # compare() / experiment.compare needs >=2 for honest LOVO
69
[/external_agent_tool_result]
```

55. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\tests\test_fusion_compare.py
[/external_agent_tool_call]
```

56. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-fusion-loaders\tests\test_fusion_compare.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

57. assistant (2026-07-02T06:11:33.308Z)

Now the audio test:

58. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\tests\test_audio_features.py
[/external_agent_tool_call]
```

59. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_result]
36      # ---------------------------------------------------- incremental-lift driver
(synthetic)
37
38      def _write_video(d, name):
39          """shuttle + person constant (no signal); audio_hit_count separates rallies (8) from
junk (0)."""
40          cands = []
41          for (s, e), pos in [((0.0, 10.0), 1), ((20.0, 30.0), 1), ((40.0, 41.0), 0), ((50.0,
51.0), 0)]:
42              shuttle = {fn: 1.0 for fn in fusion_audio.SHUTTLE_FEATURE_NAMES}
43              person = {fn: 1.0 for fn in fusion_audio.PERSON_FEATURE_NAMES}
44              audio = {fn: 0.0 for fn in af.AUDIO_FEATURE_NAMES}
45              audio["audio_hit_count"] = 8.0 if pos else 0.0
46              cands.append({"start": s, "end": e, "shuttle": shuttle, "person": person,
47                            "audio": audio, "label": pos})
48          data = {"name": name, "fps": 30.0, "frame_width": 1920.0,
49                  "candidates": cands, "gts": [[0.0, 10.0], [20.0, 30.0]]}
50          with open(os.path.join(d, f"{name}_golden_features.json"), "w", encoding="utf-8") as
f:
51              json.dump(data, f)
52
53
54      def test_audio_increment_compare(tmp_path):
55          d = str(tmp_path)
56          for k in range(6):
57              _write_video(d, f"v{k}")
```

```
58          videos = fusion_audio.load_videos_with_audio(d)
59          assert len(videos) == 6
60          res = fusion_audio.compare_increment(videos, iou=0.5)
61          assert res["variant_b"]["honest_lovo"] is True
62          assert res["delta"]["f1"] > 0.0          # audio adds separating signal on top of
constant shuttle+person
63          assert "audio" in fusion_audio.format_result(res, 0.5).lower()
64
65
66      def test_load_videos_requires_audio(tmp_path):
67          data = {"name": "x", "fps": 30.0, "frame_width": 1920.0,
68                  "candidates": [{"start": 0.0, "end": 1.0,
69                                  "shuttle": {fn: 0.0 for fn in
fusion_audio.SHUTTLE_FEATURE_NAMES},
70                                  "person": {fn: 0.0 for fn in
fusion_audio.PERSON_FEATURE_NAMES},
71                                  "label": 0}],
72                  "gts": [[0.0, 1.0]]}
73          with open(os.path.join(str(tmp_path), "x_golden_features.json"), "w",
encoding="utf-8") as f:
74              json.dump(data, f)
75          with pytest.raises(SystemExit):
76              fusion_audio.load_videos_with_audio(str(tmp_path))
77
[/external_agent_tool_result]
```

60. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-fusion-loaders\tests\test_audio_features.py
[/external_agent_tool_call]
```

61. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-fusion-loaders\tests\test_audio_features.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

62. assistant (2026-07-02T06:11:33.308Z)

Let me verify `VideoCandidates` exposes `.name`, then run the targeted tests in the worktree.

63. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Bash]
description: Confirm VideoCandidates.name attribute
command: cd "../bhi-fusion-loaders" && grep -n "class VideoCandidates\|name"
backend/eval/experiment.py | head -8
[/external_agent_tool_call]
```

64. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_result]
58:logger = logging.getLogger(__name__)
100:      - ``feature_names`` ? the persisted features the learned filter consumes. When set
107:    name: str
114:    feature_names: Optional[List[str]] = None
125:        return self.classifier_model is not None or bool(self.feature_names)
129:            "name": self.name,
136:            "feature_names": self.feature_names,
158:CONTROL = Variant(name="control")
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

65. assistant (2026-07-02T06:11:33.308Z)

```
[external_agent_tool_call: Bash]
description: Inspect VideoCandidates dataclass
command: cd "../bhi-fusion-loaders" && grep -n "class VideoCandidates"
backend/eval/experiment.py; sed -n '/class VideoCandidates/,/^class \|^def /p'
backend/eval/experiment.py | head -20
[/external_agent_tool_call]
```

66. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
165:class VideoCandidates:
class VideoCandidates:
    """One video's persisted detection, ready for offline re-scoring.

    ``intervals`` are the candidate windows; ``features`` is column-major
    (``feature_name -> per-candidate value``, each list aligned to ``intervals``);
    ``gts`` are golden labels (None for new/unlabeled footage). Build one from the DB
    with `load_video_candidates`, or directly in tests.
    """

    name: str
    intervals: List[Interval]
    features: Dict[str, List[float]] = field(default_factory=dict)
    gts: Optional[List[Interval]] = None


def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

67. assistant (2026-07-02T06:11:33.309Z)

`VideoCandidates.name` exists. Now run the two targeted test files in the worktree:

68. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_call: Bash]
description: Run targeted fusion test files
command: cd "../bhi-fusion-loaders" && python -m pytest tests/test_fusion_compare.py
tests/test_audio_features.py -q 2>&1 | tail -25
[/external_agent_tool_call]
```

69. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
..............                                                           [100%]
14 passed in 4.09s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

70. assistant (2026-07-02T06:11:33.309Z)

All 14 pass (was 12). Now the full verification gate. Let me check the repo's CI config for the exact lint/type/coverage commands.

71. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_call: Bash]
description: Find coverage floor and CI lint/type commands
command: cd "../bhi-fusion-loaders" && echo "=== pytest/cov config ==="; grep -rn "cov-fail-
under\|--cov\|addopts" pyproject.toml setup.cfg pytest.ini tox.ini 2>/dev/null | head; echo "===
mypy targets in CI ==="; grep -rn "mypy\|ruff" .github/workflows/*.yml 2>/dev/null | head
```

[/external_agent_tool_call]

72. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
=== pytest/cov config ===
=== mypy targets in CI ===
16:  # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
17:  # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
28:      - name: Install ruff
29:        run: python -m pip install ruff
31:        run: ruff check . --output-format github
33:  # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
35:  # up. Typed core deps are installed so mypy sees their real signatures
45:      - name: Install mypy + typed core deps
46:        run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-genai
numpy
48:        run: mypy backend training
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

73. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_call: Bash]
description: Inspect run_all_tests.py and coverage floor
command: cd "../bhi-fusion-loaders" && echo "=== run_all_tests.py head ==="; sed -n '1,60p'
run_all_tests.py | grep -n "cov\|pytest\|fail-under\|main\|def " | head -20; echo "=== cov-fail-
under anywhere ==="; grep -rn "cov-fail-under\|cov_fail\|fail_under" . --include=*.py
--include=*.ini --include=*.cfg --include=*.toml --include=*.yml 2>/dev/null | head
[/external_agent_tool_call]
```

74. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
=== run_all_tests.py head ===
4:A single, discoverable entry point so you (or a reviewer / CI) can verify the
8:    python run_all_tests.py -v          # verbose (passthrough to pytest)
10:    python run_all_tests.py --cov=backend --cov-report=term-missing   # coverage
12:Anything after the script name is passed straight through to pytest.
18:  package isn't installed (they use ``pytest.importorskip``). To exercise them too:
20:- Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
29:def _installed(module: str) -> bool:
36:def main() -> int:
41:    cmd = [sys.executable, "-m", "pytest", *args]
53:    print(" RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
{rc}"))
58:if __name__ == "__main__":
59:    raise SystemExit(main())
=== cov-fail-under anywhere ===
./.github/workflows/ci.yml:97:        run: python run_all_tests.py --cov=backend --cov=training
--cov-report=term --cov-fail-under=70
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

75. assistant (2026-07-02T06:11:33.309Z)

CI gate is `--cov-fail-under=70` on `backend` + `training`. Let me run the full suite with coverage, plus ruff and mypy.

76. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_call: Bash]
description: Run full suite with coverage gate
command: cd "../bhi-fusion-loaders" && python run_all_tests.py --cov=backend --cov=training
```

```
--cov-report=term --cov-fail-under=70 2>&1 | tail -30
[/external_agent_tool_call]
```

77. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
backend\providers\__init__.py                            16      1    94%
backend\providers\base.py                                 9      1    89%
backend\providers\gemini.py                              98     19    81%
backend\reporting\__init__.py                            26      2    92%
backend\reporting\harvest.py                            111      8    93%
backend\reporting\run_signals.py                         14      0   100%
backend\results.py                                       22      0   100%
backend\sports\__init__.py                               17      1    94%
backend\sports\badminton.py                              13      1    92%
backend\sports\base.py                                   12      3    75%
backend\tools\__init__.py                                 0      0   100%
backend\tools\export.py                                  75      2    97%
backend\tools\rally_slicer.py                           104     47    55%
backend\utils\__init__.py                                 0      0   100%
backend\utils\geometry.py                                29      0   100%
backend\utils\hashing.py                                  9      0   100%
backend\utils\logging_setup.py                           11      0   100%
backend\utils\paths.py                                   27      3    89%
backend\utils\run_telemetry.py                          126     12    90%
backend\utils\video.py                                   25      3    88%
training\__init__.py                                      0      0   100%
training\gen0\__init__.py                                 0      0   100%
training\gen0\harness.py                                108     11    90%
----------------------------------------------------------------------
TOTAL                                                  7992   1546    81%
Required test coverage of 70% reached. Total coverage: 80.66%
846 passed, 7 skipped, 7 warnings in 103.74s (0:01:43)
========================================================================
 RESULT: PASS - all tests green
========================================================================
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

78. assistant (2026-07-02T06:11:33.309Z)

Full suite green: **846 passed, 7 skipped**, coverage **80.66%** (?70). Now ruff + mypy:

79. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_call: Bash]
description: Run ruff and mypy gates
command: cd "../bhi-fusion-loaders" && echo "=== RUFF ===" && ruff check . 2>&1 | tail -15; echo
"=== MYPY ===" && mypy backend training 2>&1 | tail -15
[/external_agent_tool_call]
```

80. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
=== RUFF ===
All checks passed!
=== MYPY ===
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs  [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
```

```
checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 104 source files
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

81. assistant (2026-07-02T06:11:33.309Z)

```
All gates green ? suite 846 passed, coverage 80.66%, ruff clean, mypy clean (only informational
notes). Let me review the complete diff before committing.
```

82. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_call: Bash]
description: Review the full diff
command: cd "../bhi-fusion-loaders" && git diff --stat && echo "=== FULL DIFF ===" && git diff
[/external_agent_tool_call]
```

83. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
 backend/eval/fusion_audio.py   | 21 +++++++++++++++++----
 backend/eval/fusion_compare.py | 35 ++++++++++++++++++++++++++++++++---
 tests/test_audio_features.py   | 26 ++++++++++++++++++++++++++
 tests/test_fusion_compare.py   | 25 +++++++++++++++++++++++++
 4 files changed, 100 insertions(+), 7 deletions(-)
=== FULL DIFF ===
diff --git a/backend/eval/fusion_audio.py b/backend/eval/fusion_audio.py
index 8290614..d59dd46 100644
--- a/backend/eval/fusion_audio.py
+++ b/backend/eval/fusion_audio.py
@@ -15,15 +15,18 @@ from __future__ import annotations
 import argparse
 import glob
 import json
+import logging
 import os
 from typing import Any, Dict, List

 from backend.eval import experiment
-from backend.eval.fusion_compare import delta_stats
+from backend.eval.fusion_compare import _is_fusion_schema, delta_stats
 from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES,
_derive_video_path
 from backend.pipeline.detectors import audio_features as af
 from backend.pipeline.audio.hit_detector import detect_hits_in_video

+logger = logging.getLogger(__name__)
+

 def augment(manifest_path: str, features_dir: str, k: float = 2.5) -> List[str]:
     """Add the 3 audio features to each existing ``<name>_golden_features.json`` window."""
@@ -50,14 +53,24 @@ def augment(manifest_path: str, features_dir: str, k: float = 2.5) ->
List[str]:


 def load_videos_with_audio(features_dir: str) -> List[experiment.VideoCandidates]:
-    """Load the feature JSONs into VideoCandidates including the audio columns (requires
-    `augment` to have run ? raises if a JSON lacks audio)."""
+    """Load the FULL-FUSION-SCHEMA feature JSONs into VideoCandidates including the audio
+    columns (requires `augment` to have run ? raises if a fusion JSON lacks audio).
+
+    Leaner velocity-windowing-only files (no ``shuttle``/``person`` cue dicts) are SKIPPED, not
+    crashed on, so a mixed corpus loads the fusion subset cleanly (see ``_is_fusion_schema``).
+    The skip is checked BEFORE the audio-presence guard, so a genuine fusion file that simply
```

```
+    hasn't been ``--augment``-ed still raises rather than being silently dropped."""
     videos: List[experiment.VideoCandidates] = []
     for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
         with open(path, encoding="utf-8") as f:
             d = json.load(f)
         cands = d["candidates"]
-        if cands and "audio" not in cands[0]:
+        if not _is_fusion_schema(cands):
+            logger.info("fusion_audio.load_videos_with_audio: skipping %s ? windowing-only "
+                        "schema (no shuttle/person cue dicts), not a fusion-feature file",
+                        os.path.basename(path))
+            continue
+        if "audio" not in cands[0]:
             raise SystemExit(f"{path} has no audio features ? run `--augment` first")
         intervals = [(float(c["start"]), float(c["end"])) for c in cands]
         features: Dict[str, List[float]] = {}
diff --git a/backend/eval/fusion_compare.py b/backend/eval/fusion_compare.py
index 463ec7e..9372ba0 100644
--- a/backend/eval/fusion_compare.py
+++ b/backend/eval/fusion_compare.py
@@ -14,13 +14,16 @@ from __future__ import annotations
 import argparse
 import glob
 import json
+import logging
 import os
-from typing import Any, Dict, List
+from typing import Any, Dict, List, Sequence

 from backend.eval import experiment
 from backend.eval.eval_stats import paired_delta_ci
 from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES

+logger = logging.getLogger(__name__)
+

 def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
     """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign test
@@ -32,14 +35,40 @@ def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
     return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])


+def _is_fusion_schema...
[/external_agent_tool_result]
```

84. assistant (2026-07-02T06:11:33.309Z)

The diff is clean. Let me prove the regression tests genuinely catch the bug ? temporarily stash only the production fixes and confirm the new tests fail against the old loaders.

85. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_call: Bash]
description: Confirm new tests fail without the fix
command: cd "../bhi-fusion-loaders" && git stash push backend/eval/fusion_compare.py
backend/eval/fusion_audio.py >/dev/null 2>&1 && echo "--- production fix stashed; running new
tests against OLD loaders ---" && python -m pytest
"tests/test_fusion_compare.py::test_load_videos_skips_windowing_only_in_mixed_corpus"
"tests/test_audio_features.py::test_load_videos_with_audio_skips_windowing_only_in_mixed_corpus"
-q 2>&1 | tail -20; echo "--- restoring production fix ---"; git stash pop >/dev/null 2>&1 &&
echo "restored"
[/external_agent_tool_call]
```

86. assistant (2026-07-02T06:11:33.309Z)

```
[external_agent_tool_result]
--- production fix stashed; running new tests against OLD loaders ---

features_dir = 'C:\\Users\\avidu\\AppData\\Local\\Temp\\pytest-of-
avidu\\pytest-490\\test_load_videos_with_audio_sk0'

    def load_videos_with_audio(features_dir: str) -> List[experiment.VideoCandidates]:
        """Load the feature JSONs into VideoCandidates including the audio columns (requires
        `augment` to have run ? raises if a JSON lacks audio)."""
        videos: List[experiment.VideoCandidates] = []
        for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
            with open(path, encoding="utf-8") as f:
                d = json.load(f)
            cands = d["candidates"]
            if cands and "audio" not in cands[0]:
>               raise SystemExit(f"{path} has no audio features ? run `--augment` first")
E               SystemExit: C:\Users\avidu\AppData\Local\Temp\pytest-of-
avidu\pytest-490\test_load_videos_with_audio_sk0\window0_golden_features.json has no audio
features ? run `--augment` first

backend\eval\fusion_audio.py:61: SystemExit
=========================== short test summary info ===========================
FAILED tests/test_fusion_compare.py::test_load_videos_skips_windowing_only_in_mixed_corpus
FAILED
tests/test_audio_features.py::test_load_videos_with_audio_skips_windowing_only_in_mixed_corpus
2 failed in 3.20s
--- restoring production fix ---
restored
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

87. assistant (2026-07-02T06:11:33.310Z)

Both regression tests fail without the fix and pass with it ? genuinely validated (the audio one even exposes the misleading "run --augment first" error the old code raised on windowing-only files). Now commit, following the shared-checkout safety protocol: re-verify HEAD, stage explicit paths only.

88. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_call: Bash]
description: Re-verify branch state before commit
command: cd "../bhi-fusion-loaders" && echo "=== branch ===" && git branch --show-current &&
echo "=== HEAD ===" && git log --oneline -1 && echo "=== commits vs origin/master (should be
empty pre-commit) ===" && git log --oneline origin/master..HEAD && echo "=== status ===" && git
status --short
[/external_agent_tool_call]
```

89. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_result]
=== branch ===
fix/fusion-loaders-mixed-schema
=== HEAD ===
be39a70 docs: rally-detection quality report ? technical-paper checkpoint (2026-06-17) (#206)
=== commits vs origin/master (should be empty pre-commit) ===
=== status ===
 M backend/eval/fusion_audio.py
 M backend/eval/fusion_compare.py
 M tests/test_audio_features.py
 M tests/test_fusion_compare.py
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

90. assistant (2026-07-02T06:11:33.310Z)

Clean state ? HEAD is at `origin/master`, only my 4 files modified. Staging explicit paths and committing:

91. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_call: Bash]
description: Stage explicit paths and commit
command: cd "../bhi-fusion-loaders" && git add backend/eval/fusion_compare.py
backend/eval/fusion_audio.py tests/test_fusion_compare.py tests/test_audio_features.py && git
commit -q -F - <<'EOF' && git log --oneline -1
fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus

`fusion_compare.load_videos` and `fusion_audio.load_videos_with_audio`
read `c["shuttle"]`/`c["person"]` (and audio) unconditionally on every
candidate, so they `KeyError: 'shuttle'` on a MIXED `output/` corpus where
some `*_golden_features.json` files carry the leaner velocity-windowing /
distill schema (start/end + velocity stats, no per-cue dicts). Same latent
bug just fixed in `ablation.py::load_videos`.

Mirror that fix: a local `_is_fusion_schema(candidates)` predicate
(full-fusion iff every candidate has `shuttle`+`person`; audio optional)
gates each loader ? windowing-only files are skipped with an info log
instead of crashing, so a mixed corpus loads the fusion subset cleanly.
In the audio loader the skip is checked BEFORE the audio-presence guard,
so a genuine fusion file that simply hasn't been `--augment`-ed still
raises rather than being silently dropped.

The predicate is duplicated (not imported from `ablation.py`): that fix is
on an unmerged branch, so importing would stack this PR on it. `fusion_audio`
imports the copy from `fusion_compare` (already a dependency ? no awkward
import).

Latent today: the existing tests only exercise synthetic tmp_path corpora,
never the real `output/`. Adds a per-loader regression test (mixed corpus
loads only the fusion files); both verified to fail without the fix.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>
EOF
[/external_agent_tool_call]
```

92. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_result]
25a5a78 fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

93. assistant (2026-07-02T06:11:33.310Z)

Committed. Let me verify the commit is clean (single commit vs origin/master, LF line endings) and push, then open the PR.

94. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_call: Bash]
description: Verify single clean LF commit
command: cd "../bhi-fusion-loaders" && echo "=== commits vs origin/master (should be ONLY mine)
===" && git log --oneline origin/master..HEAD && echo "=== CRLF check on committed blobs (should
be i/lf) ===" && git ls-files --eol backend/eval/fusion_compare.py backend/eval/fusion_audio.py
tests/test_fusion_compare.py tests/test_audio_features.py && echo "=== any CR in committed diff?
===" && (git show HEAD | grep -c $'\r' && echo "HAS CR (bad)" || echo "0 CR ? LF clean")
[/external_agent_tool_call]
```

95. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_result]
=== commits vs origin/master (should be ONLY mine) ===
25a5a78 fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus
=== CRLF check on committed blobs (should be i/lf) ===
i/lf    w/crlf  attr/                   backend/eval/fusion_audio.py
i/lf    w/crlf  attr/                   backend/eval/fusion_compare.py
i/lf    w/crlf  attr/                   tests/test_audio_features.py
i/lf    w/crlf  attr/                   tests/test_fusion_compare.py
=== any CR in committed diff? ===
0
0 CR ? LF clean
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

96. assistant (2026-07-02T06:11:33.310Z)

Single clean LF commit, only mine vs `origin/master`. Pushing and opening the PR:

97. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_call: Bash]
description: Push the branch to origin
command: cd "../bhi-fusion-loaders" && git push -u origin fix/fusion-loaders-mixed-schema 2>&1 |
tail -8
[/external_agent_tool_call]
```

98. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_result]
remote:
remote: Create a pull request for 'fix/fusion-loaders-mixed-schema' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/fusion-
loaders-mixed-schema
remote:
branch 'fix/fusion-loaders-mixed-schema' set up to track 'origin/fix/fusion-loaders-mixed-
schema'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      fix/fusion-loaders-mixed-schema -> fix/fusion-loaders-mixed-schema
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

99. assistant (2026-07-02T06:11:33.310Z)

```
[external_agent_tool_call: Bash]
description: Open the pull request
command: cd "../bhi-fusion-loaders" && gh pr create --base master --head fix/fusion-loaders-
mixed-schema --title "fix(eval): fusion loaders skip windowing-only files on a mixed golden
corpus" --body "$(cat <<'EOF'
```

What

`fusion_compare.load_videos` and `fusion_audio.load_videos_with_audio` read `c["shuttle"]` /
`c["person"]` (and audio) **unconditionally** on every candidate, so they crash with `KeyError:
'shuttle'` on a **mixed** golden corpus where some `output/*_golden_features.json` files carry
the leaner velocity-windowing / distill schema (`active_frame_count`, `chunk_index`, `start`,
`end`, `mean_velocity`, `peak_velocity`, `track_id_count` ? no per-cue dicts).

This is the **same latent bug** just fixed in `backend/eval/ablation.py::load_videos`. Running
either CLI on the real n=11 corpus would `KeyError`:
- `python -m backend.eval.fusion_compare --features-dir output`
- `python -m backend.eval.fusion_audio --features-dir output --compare`

It is **latent, not currently red**: `tests/test_fusion_compare.py` and `tests/test_audio_features.py` only exercise synthetic `tmp_path` corpora, never the real `output/`.

Fix

Mirror the ablation fix ? a `_is_fusion_schema(candidates)` predicate (full-fusion iff every candidate has `shuttle` + `person`; `audio` optional) gates each loader. Windowing-only files are **skipped with an info log** instead of crashing, so a mixed corpus loads the fusion subset cleanly.

- In the **audio** loader the skip is checked **before** the audio-presence guard, so a genuine fusion file that simply hasn't been `--augment`-ed still raises (`run --augment first`) rather than being silently dropped. `tests/test_audio_features.py::test_load_videos_requires_audio` still passes.
- The predicate is **duplicated** rather than imported from `ablation.py`: that fix is on an unmerged branch, so importing would **stack** this PR on it. `fusion_audio` imports the copy from `fusion_compare` (already a dependency ? no awkward import). A note in the docstring records the intentional duplication for a future consolidation once both land.

Tests

Adds one regression test per loader (`test_load_videos_skips_windowing_only_in_mixed_corpus` / `test_load_videos_with_audio_skips_windowing_only_in_mixed_corpus`): a mixed corpus loads **only** the fusion files. Both **verified to fail without the fix** (KeyError / misleading `SystemExit`) and pass with it. Offline / fully mocked.

Verification (local)

- `python run_all_tests.py --cov=backend --cov=training --cov-fail-under=70` ? **846 passed, 7 skipped**, coverage **80.66%** (?70 floor, no regression).
- `ruff check .` ? clean.
- `mypy backend training` ? clean (no issues in 104 files).

? Generated with [Claude Code](https://claude.com/claude-code)
EOF
)" 2>&1 | tail -5
[/external_agent_tool_call]

100. assistant (2026-07-02T06:11:33.310Z)

[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/210
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

101. assistant (2026-07-02T06:11:33.310Z)

PR opened: [#210](https://github.com/avidullu/badminton-highlight-indexer/pull/210). Now clean up the isolated worktree (branch is pushed, so it's safe to remove):

102. assistant (2026-07-02T06:11:33.310Z)

[external_agent_tool_call: Bash]
description: Remove the isolated worktree
command: git worktree remove "../bhi-fusion-loaders" 2>&1 && echo "worktree removed" && git worktree list
[/external_agent_tool_call]

103. assistant (2026-07-02T06:11:33.311Z)

[external_agent_tool_result]
worktree removed
C:/Users/avidu/Projects/badminton-highlight-indexer

f5f5e52 [docs/golden-regression-fixtures-plan]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/ablation-schema-fix
59ae347 [fix/ablation-litmus-mixed-schema]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
8736e2d [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-golden
c33cf1d [docs/golden-videos-tracker]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality
877ab1f [docs/rally-quality-research]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-today
ce7eb47 [docs/quality-iterations-realfootage]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-noai
3ea7ad4 [feat/local-no-ai-segmenter]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/pr-stitch
a6adcc5 [fix/runner-stitch-watermark]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/r2-impl
a450c53 [feat/r2-tolerance-metric]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/r4-endrule
e248d2d [feat/r4-inactivity-end]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-eval
9e03e5a [feat/tier0-rally-seg-eval]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-netcross
effd920 [feat/w2-netcross-gate]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]
[/external_agent_tool_result]

104. assistant (2026-07-02T06:11:33.311Z)

Done. PR [#210](https://github.com/avidullu/badminton-highlight-indexer/pull/210) is open.

What I did

Fixed the same latent `KeyError: 'shuttle'` bug in both fusion golden-feature loaders that was
just fixed in `ablation.py::load_videos`:

- **backend/eval/fusion_compare.py** ? added a local
`_is_fusion_schema(candidates)` predicate (full-fusion iff every candidate has
`shuttle`+`person`; audio optional) and a module logger; `load_videos` now skips windowing-only
files with an info log instead of crashing.
- **backend/eval/fusion_audio.py** ? imports that predicate from
`fusion_compare` (already a dependency ? `delta_stats` came from there, so no awkward import)
and gates `load_videos_with_audio`. The skip is checked **before** the audio-presence guard, so
a genuine fusion file that just hasn't been `--augment`-ed still raises, but windowing-only
files are dropped cleanly.
- **One regression test per loader** asserting a mixed corpus loads only the fusion files.

Key decisions
- **Duplicated the predicate rather than importing from `ablation.py`** ? that fix is on an
unmerged branch (`fix/ablation-litmus-mixed-schema`), so importing would stack this PR on it.
The predicate's a one-liner; the docstring records the intentional duplication for a future
consolidation once both land. This honored "do not stack on the ablation fix."

Verification (local, in an isolated worktree off `origin/master`)
- `python run_all_tests.py --cov=backend --cov=training --cov-fail-under=70` ? **846 passed, 7
skipped**, coverage **80.66%** (?70 floor, no regression)

- `ruff check .` ? clean
- `mypy backend training` ? clean (no issues in 104 files)
- **Both new tests verified to fail without the fix** (KeyError / the misleading `SystemExit`) and pass with it ? they're genuine, not vacuous.

Branch was cut from `origin/master`, HEAD re-verified before commit (`origin/master..HEAD` = my single commit only), explicit paths staged, and the worktree removed after pushing ? per the shared-checkout safety protocol.

One thing worth flagging: when `fix/ablation-litmus-mixed-schema` and this both merge, there'll be two copies of the `_is_fusion_schema` one-liner. A small follow-up could consolidate them into a shared `backend/eval` helper ? but that's intentionally out of scope here to avoid the stacking dependency.

105. user (2026-07-02T06:11:33.311Z)

I think both are merged, can you do the small follow up as a diligence cleanup?

106. assistant (2026-07-02T06:11:33.311Z)

Let me scout the current state first ? confirm both fixes are merged, pull master, and map all copies/usages of `_is_fusion_schema` before deciding the shared location.

107. assistant (2026-07-02T06:11:33.311Z)

[external_agent_tool_call: Bash]
description: Verify both fixes merged into origin/master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git fetch origin --quiet && echo "=== origin/master tip ===" && git log --oneline -5 origin/master && echo && echo "=== is ablation fix merged? ===" && git branch -a --contains $(git rev-parse fix/ablation-litmus-mixed-schema 2>/dev/null) 2>/dev/null | grep -i "origin/master" && echo "ablation MERGED" || echo "ablation NOT merged (by contains)" && echo && echo "=== is #210 (fusion loaders) merged? ===" && git branch -a --contains $(git rev-parse fix/fusion-loaders-mixed-schema 2>/dev/null) 2>/dev/null | grep -i "origin/master" && echo "fusion MERGED" || echo "fusion NOT merged (by contains)"
[/external_agent_tool_call]

108. assistant (2026-07-02T06:11:33.311Z)

[external_agent_tool_result]
=== origin/master tip ===
f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
db96ddc docs: capture external due-diligence review inputs (research, pick-up-later) (#207)
af5b3e6 docs(rally): tracked doc for closing the detection accuracy gap (golden-driven, per-video) (#211)
55e4c89 docs(archive): LOVO threshold re-sweep at n=13 ? milestone thresholds hold (#214)
bd33085 fix(scripts): unwrap segmenter Result before len() (motion crashes the default eval path) (#213)

=== is ablation fix merged? ===
ablation NOT merged (by contains)

=== is #210 (fusion loaders) merged? ===
fusion NOT merged (by contains)
[/external_agent_tool_result]

109. assistant (2026-07-02T06:11:33.311Z)

Squash-merges, so `--contains` won't match ? #210 is clearly at the tip (f5f339a). Let me verify by content: check whether `_is_fusion_schema` exists in both `ablation.py` and `fusion_compare.py` on `origin/master`.

110. assistant (2026-07-02T06:11:33.311Z)

```
[external_agent_tool_call: Bash]
description: Find _is_fusion_schema across origin/master
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== _is_fusion_schema
definitions/usages on origin/master ===" && git grep -n "_is_fusion_schema" origin/master --
'backend/**' 'tests/**' 2>/dev/null; echo; echo "=== ablation fix merged via PR? search log ==="
&& git log --oneline origin/master | grep -i
"ablation.*litmus\|ablation.*mixed\|ablation.*schema" | head -3
[/external_agent_tool_call]
```

111. assistant (2026-07-02T06:11:33.311Z)

```
[external_agent_tool_result]
=== _is_fusion_schema definitions/usages on origin/master ===
origin/master:backend/eval/ablation.py:386:def _is_fusion_schema(candidates: Sequence[Dict[str,
Any]]) -> bool:
origin/master:backend/eval/ablation.py:406:    ablatable subset cleanly (see
``_is_fusion_schema``)."""
origin/master:backend/eval/ablation.py:412:        if not _is_fusion_schema(cands):
origin/master:backend/eval/fusion_audio.py:23:from backend.eval.fusion_compare import
_is_fusion_schema, delta_stats
origin/master:backend/eval/fusion_audio.py:60:    crashed on, so a mixed corpus loads the fusion
subset cleanly (see ``_is_fusion_schema``).
origin/master:backend/eval/fusion_audio.py:68:        if not _is_fusion_schema(cands):
origin/master:backend/eval/fusion_compare.py:38:def _is_fusion_schema(candidates:
Sequence[Dict[str, Any]]) -> bool:
origin/master:backend/eval/fusion_compare.py:61:    fusion subset cleanly (see
``_is_fusion_schema``)."""
origin/master:backend/eval/fusion_compare.py:67:        if not _is_fusion_schema(cands):
origin/master:tests/test_ablation.py:373:def
test_is_fusion_schema_distinguishes_lean_and_full():
origin/master:tests/test_ablation.py:379:    assert ablation._is_fusion_schema(full) is True
origin/master:tests/test_ablation.py:380:    assert ablation._is_fusion_schema(no_audio) is True
origin/master:tests/test_ablation.py:381:    assert ablation._is_fusion_schema(lean) is False
origin/master:tests/test_ablation.py:382:    assert ablation._is_fusion_schema([]) is False
origin/master:tests/test_ablation.py:420:        if ablation._is_fusion_schema(cands):

=== ablation fix merged via PR? search log ===
4bd9d2d fix(eval): ablation litmus tolerates a mixed-schema golden corpus (#209)
[/external_agent_tool_result]
```

112. assistant (2026-07-02T06:11:33.311Z)

Both are merged: ablation fix is PR #209 (`_is_fusion_schema` in `ablation.py` + tested in `test_ablation.py`), fusion fix is #210 (`_is_fusion_schema` in `fusion_compare.py`, imported by `fusion_audio.py`). So there are now **two identical definitions**. Let me map the import graph and pick a non-circular shared home.

113. assistant (2026-07-02T06:11:33.311Z)

```
[external_agent_tool_call: Bash]
description: List eval modules and inspect fusion_golden imports
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== backend/eval
modules ===" && ls backend/eval/*.py && echo && echo "=== fusion_golden.py imports (potential
shared home ? must NOT import ablation/fusion_compare/fusion_audio) ===" && git show
origin/master:backend/eval/fusion_golden.py | grep -n "^import\|^from\|^def
\|^SHUTTLE\|^PERSON\|FEATURE_NAMES" | head -40
[/external_agent_tool_call]
```

114. assistant (2026-07-02T06:11:33.311Z)

```
[external_agent_tool_result]
=== backend/eval modules ===
backend/eval/__init__.py
backend/eval/ablation.py
```

```
backend/eval/annotations.py
backend/eval/batch.py
backend/eval/calibrate_local.py
backend/eval/calibrate_wasb.py
backend/eval/calibration.py
backend/eval/classifier.py
backend/eval/distill_local.py
backend/eval/eval_partial_labels.py
backend/eval/eval_stats.py
backend/eval/experiment.py
backend/eval/export_wasb_segments.py
backend/eval/fusion_audio.py
backend/eval/fusion_compare.py
backend/eval/fusion_golden.py
backend/eval/gemini_refine.py
backend/eval/golden_fixtures.py
backend/eval/gt_loader.py
backend/eval/harness.py
backend/eval/metrics.py
backend/eval/per_user_eval.py
backend/eval/per_user_gate.py
backend/eval/per_user_regression.py
backend/eval/promotion.py
backend/eval/rally_seg_eval.py
backend/eval/rally_seq_proto.py
backend/eval/regression.py
backend/eval/score_calibration.py
backend/eval/segmentation_metrics.py
backend/eval/serve_contrast.py
backend/eval/served_gate_a.py
backend/eval/splits.py
backend/eval/tolerance_metrics.py
backend/eval/training.py
backend/eval/windowing.py

=== fusion_golden.py imports (potential shared home ? must NOT import
ablation/fusion_compare/fusion_audio) ===
17:from __future__ import annotations
19:import argparse
20:import json
21:import os
22:import time
23:from typing import Any, Dict, List, Optional
25:from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
26:from backend.pipeline.detectors import rally_gate
27:from backend.pipeline.detectors import fusion_features as ff
28:from backend.eval import distill_local
29:from backend.eval.calibrate_wasb import load_golden_gts
30:from backend.eval.training import label_candidates
31:from backend.utils.run_telemetry import RunTelemetry
34:SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
35:# The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
36:PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
39:def _derive_video_path(spec: Dict[str, Any]) -> str:
50:def extract_video(spec: Dict[str, Any], detector: Optional[Any] = None,
118:            "shuttle": {k: float(v) for k, v in zip(SHUTTLE_FEATURE_NAMES, xs)},
129:def run(manifest_path: str, out_dir: str, frame_skip: int = 3, iou: float = 0.5,
180:def main() -> None:
[/external_agent_tool_result]
```

115. assistant (2026-07-02T06:11:33.311Z)

`fusion_golden.py` is heavy (pulls in `TrackNetRunner`, `rally_gate`, `fusion_features`) ? wrong home for a pure dict-key predicate; it'd drag heavy deps into `ablation.py`. Let me check the lighter shared candidate `experiment.py` (already imported by all three consumers) and confirm `ablation.py`'s current imports + no circular risk.

116. assistant (2026-07-02T06:11:33.312Z)

[external_agent_tool_call: Bash]
description: Check ablation and experiment import graphs
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== ablation.py imports ===" && git show origin/master:backend/eval/ablation.py | grep -n "^import\|^from" | head -30 && echo && echo "=== experiment.py imports (is it light? circular?) ===" && git show origin/master:backend/eval/experiment.py | grep -n "^import\|^from" | head -30
[/external_agent_tool_call]

117. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_result]
=== ablation.py imports ===
46:from __future__ import annotations
48:import argparse
49:import glob
50:import json
51:import logging
52:import os
53:from dataclasses import dataclass, field
54:from typing import Any, Dict, List, Optional, Sequence, Tuple
56:from backend.eval import experiment
57:from backend.eval.experiment import Variant, VideoCandidates

=== experiment.py imports (is it light? circular?) ===
38:from __future__ import annotations
40:import json
41:import logging
42:import os
43:from dataclasses import dataclass, field
44:from datetime import datetime, timezone
45:from hashlib import sha1
46:from typing import Any, Callable, Dict, List, Optional, Sequence
48:from backend.eval.calibration import pct_improvement
49:from backend.eval.classifier import LogisticRegression
50:from backend.eval.gemini_refine import merge_overlaps
51:from backend.eval.metrics import Interval, match_intervals, score
52:from backend.eval.regression import gt_hash
53:from backend.eval.training import label_candidates
54:from backend.eval import eval_stats as _stats          # C1: CIs + paired sign test
55:from backend.eval import segmentation_metrics as _segm   # C4: F1@k + segment-count ratio
56:from backend.eval.score_calibration import fit_isotonic, fit_platt   # C2: calibrate cue
scores
[/external_agent_tool_result]
```

118. assistant (2026-07-02T06:11:33.312Z)

All three consumers (`ablation`, `fusion_compare`, `fusion_audio`) already import `experiment`, and `experiment.py` imports none of them ? so it's a **circular-safe, zero-new-edge** home, and it owns `VideoCandidates` (the type these JSONs deserialize into). That's the right shared spot. Let me look at where to place it and how the existing predicate test is structured.

119. assistant (2026-07-02T06:11:33.312Z)

[external_agent_tool_call: Bash]
description: Check for test_experiment.py and experiment.py function inventory
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && echo "=== test_experiment.py exists? ===" && ls tests/test_experiment.py 2>/dev/null || echo "NO test_experiment.py" && echo && echo "=== experiment.py def inventory + does it touch *_golden_features.json / load_video_candidates? ===" && git show origin/master:backend/eval/experiment.py | grep -n "^def \|^class \|golden_features\|def

```
load_video_candidates\|^    name:\|candidates" | head -40
[/external_agent_tool_call]
```

120. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_result]
=== test_experiment.py exists? ===
tests/test_experiment.py

=== experiment.py def inventory + does it touch *_golden_features.json / load_video_candidates?
===
10:  - `backend.eval.training`       ? `label_candidates` (y by the same IoU matcher)
35:`load_video_candidates`, which reads persisted candidates via
53:from backend.eval.training import label_candidates
67:class ExperimentError(ValueError):
77:class Variant:
107:    name: str
156:#: The pinned, frozen baseline: candidates as detected, merged, no gate, no learned
161:# ----------------------------------------------------------------- persisted candidates
165:class VideoCandidates:
171:    with `load_video_candidates`, or directly in tests.
174:    name: str
180:def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
191:            f"feature {name!r} has {len(col)} values but {vc.name} has {n} candidates")
195:def _feature_matrix(vc: VideoCandidates, feature_names: Sequence[str]) -> List[List[float]]:
200:def _gate_mask(variant: Variant, vc: VideoCandidates) -> List[bool]:
213:def predict(variant: Variant, vc: VideoCandidates,
243:def _calibrate_and_tune(variant: Variant, model: LogisticRegression,
257:            y.extend(label_candidates(vc.intervals, vc.gts or [], iou))
281:def _train(variant: Variant, videos: Sequence[VideoCandidates], iou: float) ->
LogisticRegression:
282:    """Fit the variant's classifier on the pooled candidates of ``videos`` (labels via the
290:        y.extend(label_candidates(vc.intervals, vc.gts, iou))
292:        raise ExperimentError("cannot train a learned variant on zero candidates")
296:def evaluate_variant(videos: Sequence[VideoCandidates], variant: Variant,
359:def _ci_block(eval_v: Dict[str, Any]) -> Dict[str, Any]:
364:def _segmentation_block(videos: Sequence[VideoCandidates], eval_v: Dict[str, Any]) ->
Dict[str, Any]:
384:def honest_summary(videos: Sequence[VideoCandidates], eval_a: Dict[str, Any],
403:def compare(videos: Sequence[VideoCandidates], variant_a: Variant, variant_b: Variant,
454:def compare_unlabeled(video: VideoCandidates, variant_a: Variant, variant_b: Variant,
504:def record_experiment(result: Dict[str, Any], path: str = DEFAULT_LEADERBOARD_PATH,
543:def load_video_candidates(db: Any, video_id: str, *, detector: Optional[str] = None,
[/external_agent_tool_result]
```

121. assistant (2026-07-02T06:11:33.312Z)

`experiment.py` owns `VideoCandidates` (line 165) and `load_video_candidates` ? the predicate belongs right after the dataclass. `tests/test_experiment.py` exists, so the canonical unit test can live there honestly. Let me set up an isolated worktree off `origin/master` and read the live files.

122. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_call: Bash]
description: Create worktree off origin/master for the consolidation
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git worktree add -b
chore/consolidate-fusion-schema-predicate "../bhi-consolidate-schema" origin/master 2>&1 | tail
-3 && echo "=== HEAD ===" && git -C "../bhi-consolidate-schema" log --oneline -1 && git -C
"../bhi-consolidate-schema" branch --show-current
[/external_agent_tool_call]
```

123. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_result]
Preparing worktree (new branch 'chore/consolidate-fusion-schema-predicate')
branch 'chore/consolidate-fusion-schema-predicate' set up to track 'origin/master'.
HEAD is now at f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
=== HEAD ===
f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
chore/consolidate-fusion-schema-predicate
[/external_agent_tool_result]
```

124. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_result]
# ---------------------------------------------------------------------------- corpus I/O


def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) -> bool:
    """True iff every candidate row carries the per-cue feature dicts the ablation scorer reads
    (``shuttle`` + ``person``; ``audio`` stays optional ? older JSONs predate it).

    The velocity-windowing / distill path persists a LEANER ``*_golden_features.json`` row
    (``start``/``end`` + velocity stats, no cue dicts) for clips that never went through fusion
    feature extraction. ``load_videos`` SKIPS those rather than crash on ``c["shuttle"]`` ? so a
    mixed corpus (some fusion files, some windowing-only) loads the ablatable subset cleanly.
    An empty candidate list is not ablatable either."""
    return bool(candidates) and all("shuttle" in c and "person" in c for c in candidates)


def load_videos(features_dir: str) -> List[VideoCandidates]:
    """Load every FULL-FUSION-SCHEMA ``*_golden_features.json`` into a ``VideoCandidates``
    (intervals + shuttle + person + audio columns + golden gts). Audio columns are optional
    (older JSONs predate the audio augment) ? a missing block defaults those columns to 0.0 so
    the loader never crashes, mirroring ``experiment._feature_column``'s tolerance.

    Leaner velocity-windowing-only files (no ``shuttle``/``person`` cue dicts ? the schema the
    R3/R4 corpus-growth path persists) are SKIPPED, not crashed on, so a mixed corpus loads the
    ablatable subset cleanly (see ``_is_fusion_schema``)."""
    videos: List[VideoCandidates] = []
    for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
        with open(path, encoding="utf-8") as f:
            d = json.load(f)
        cands = d["candidates"]
        if not _is_fusion_schema(cands):
            logger.info("ablation.load_videos: skipping %s ? windowing-only schema "
                        "(no shuttle/person cue dicts), not an ablatable fusion-feature file",
                        os.path.basename(path))
            continue
        intervals = [(float(c["start"]), float(c["end"])) for c in cands]
        features: Dict[str, List[float]] = {}
        for fn in SHUTTLE_COLUMNS:
            features[fn] = [float(c["shuttle"][fn]) for c in cands]
        for fn in PERSON_COLUMNS:
```

```
                features[fn] = [float(c["person"][fn]) for c in cands]
            for fn in AUDIO_COLUMNS:
                features[fn] = [float((c.get("audio") or {}).get(fn, 0.0)) for c in cands]
[/external_agent_tool_result]
```

125. assistant (2026-07-02T06:11:33.312Z)

Now let me read the live files in the worktree ? the current `_is_fusion_schema` defs in `ablation.py` and `fusion_compare.py`, the `experiment.py` placement site, and the existing predicate test.

126. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\ablation.py
[/external_agent_tool_call]
```

127. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_ablation.py
[/external_agent_tool_call]
```

128. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_result]
                "candidates": cands, "gts": [[0.0, 5.0]]}
        with open(os.path.join(d, f"{name}_golden_features.json"), "w", encoding="utf-8") as f:
            json.dump(data, f)


    def test_is_fusion_schema_distinguishes_lean_and_full():
        """The schema predicate keeps full-fusion rows (shuttle+person, audio optional) and rejects
        the leaner velocity-windowing rows (no cue dicts); an empty candidate list is not ablatable."""
        full = [{"start": 0.0, "end": 1.0, "shuttle": {}, "person": {}, "audio": {}}]
        no_audio = [{"start": 0.0, "end": 1.0, "shuttle": {}, "person": {}}]          # audio optional
        lean = [{"start": 0.0, "end": 1.0, "mean_velocity": 1.0, "track_id_count": 1}]
        assert ablation._is_fusion_schema(full) is True
        assert ablation._is_fusion_schema(no_audio) is True
        assert ablation._is_fusion_schema(lean) is False
        assert ablation._is_fusion_schema([]) is False


    def test_load_videos_skips_windowing_only_schema(tmp_path):
        """Mixed-corpus regression (the n=11 ``KeyError: 'shuttle'``): full-fusion files load; the
        leaner velocity-windowing files (the 5 newer real clips) are SKIPPED rather than crashed on,
        and the litmus path runs end-to-end over the filtered ablatable subset."""
        d = str(tmp_path)
        _write_learned_corpus(d, n=6)                                 # 6 full-fusion files (v0..v5)
        for name in ("mahadevpura_1", "GX010128", "Badminton_BXH_2"):  # 3 leaner real-clip names
            _write_windowing_only(d, name)
        # only the fusion-schema files load ? the leaner ones never crash the loader
        videos = ablation.load_videos(d)
        assert {v.name for v in videos} == {f"v{k}" for k in range(6)}
        # ?and the litmus path (which reads the cue columns) runs cleanly on the mixed corpus
```

```
397         report = run_ablation(videos, ablation.litmus_config())
398         assert [r.signal for r in report.rows] == ["gate_a"]
399         assert report.num_videos == 6
400
401
402     # -------------------------------------------------------------------------- the
golden litmus
403
404     GOLDEN_DIR = os.path.join(
405         os.path.dirname(os.path.dirname(os.path.abspath(__file__))), "output")
406
407
408     def _golden_present() -> bool:
409         """True iff ?6 FULL-FUSION-SCHEMA golden files exist (the litmus needs the
shuttle/person cue
410         columns). Counts only ablatable files ? leaner velocity-windowing files are skipped
by
411         ``load_videos``, so counting them would trip the litmus on an un-ablatable corpus
(the n=11
412         mixed-schema bug)."""
413         n = 0
414         for p in glob.glob(os.path.join(GOLDEN_DIR, "*_golden_features.json")):
415             try:
416                 with open(p, encoding="utf-8") as f:
417                     cands = json.load(f).get("candidates") or []
418             except (OSError, ValueError):
419                 continue
420             if ablation._is_fusion_schema(cands):
421                 n += 1
422         return n >= 6
423
424
425     @pytest.mark.skipif(not _golden_present(),
426                         reason="golden output/*_golden_features.json absent (run
fusion_golden first)")
427     def test_litmus_reproduces_gate_a():
[/external_agent_tool_result]
```

129. assistant (2026-07-02T06:11:33.312Z)

Now I have the full picture. Two `test_ablation.py` references: the canonical unit test (lines 373?382) and `_golden_present()` (line 420). Let me check `Sequence` usage (to clean up imports after removing the defs), confirm no other references exist, and inspect `test_experiment.py`'s structure.

130. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_call: Bash]
description: Check Sequence usage, all references, and test_experiment.py head
command: cd "../bhi-consolidate-schema" && echo "=== Sequence usage in ablation.py (besides the
def being removed) ===" && grep -n "Sequence" backend/eval/ablation.py; echo "=== Sequence usage
in fusion_compare.py ===" && grep -n "Sequence" backend/eval/fusion_compare.py; echo "=== ALL
_is_fusion_schema references repo-wide (sanity) ===" && grep -rn "_is_fusion_schema" backend/
tests/ scripts/ docs/ 2>/dev/null; echo "=== test_experiment.py imports/head ===" && sed -n
'1,25p' tests/test_experiment.py
[/external_agent_tool_call]
```

131. assistant (2026-07-02T06:11:33.312Z)

```
[external_agent_tool_result]
=== Sequence usage in ablation.py (besides the def being removed) ===
54:from typing import Any, Dict, List, Optional, Sequence, Tuple
211:def _ablated_variant(full: Variant, columns: Sequence[str], gate_knob: Optional[str]) ->
Variant:
```

```
248:def ablate_one(videos: Sequence[VideoCandidates], cfg: AblationConfig,
298:def _collapse_to_groups(signals: Sequence[Signal]) -> List[Signal]:
318:def run_ablation(videos: Sequence[VideoCandidates], cfg: AblationConfig) -> AblationReport:
386:def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) -> bool:
453:    def _fmt_row(cells: Sequence[str]) -> str:
=== Sequence usage in fusion_compare.py ===
19:from typing import Any, Dict, List, Sequence
38:def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) -> bool:
=== ALL _is_fusion_schema references repo-wide (sanity) ===
backend/eval/ablation.py:386:def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) ->
bool:
backend/eval/ablation.py:406:    ablatable subset cleanly (see ``_is_fusion_schema``)."""
backend/eval/ablation.py:412:        if not _is_fusion_schema(cands):
backend/eval/fusion_audio.py:23:from backend.eval.fusion_compare import _is_fusion_schema,
delta_stats
backend/eval/fusion_audio.py:60:    crashed on, so a mixed corpus loads the fusion subset
cleanly (see ``_is_fusion_schema``).
backend/eval/fusion_audio.py:68:        if not _is_fusion_schema(cands):
backend/eval/fusion_compare.py:38:def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) ->
bool:
backend/eval/fusion_compare.py:61:    fusion subset cleanly (see ``_is_fusion_schema``)."""
backend/eval/fusion_compare.py:67:        if not _is_fusion_schema(cands):
tests/test_ablation.py:373:def test_is_fusion_schema_distinguishes_lean_and_full():
tests/test_ablation.py:379:    assert ablation._is_fusion_schema(full) is True
tests/test_ablation.py:380:    assert ablation._is_fusion_schema(no_audio) is True
tests/test_ablation.py:381:    assert ablation._is_fusion_schema(lean) is False
tests/test_ablation.py:382:    assert ablation._is_fusion_schema([]) is False
tests/test_ablation.py:420:        if ablation._is_fusion_schema(cands):
=== test_experiment.py imports/head ===
"""Tests for the experiment harness (backend.eval.experiment).

Pure/offline ? synthetic candidate rows, no GPU/WSL. The load-bearing test is the
**no-regression invariant** (`test_disabled_cues_byte_identical_to_control`): a variant
that merely carries a disabled cue must predict exactly what control predicts. That is the
guarantee that lets new cues be merged freely (`EXPERIMENT_HARNESS.md` §6).
"""
import json

import pytest

from backend.eval.classifier import LogisticRegression
from backend.eval.experiment import (
    CONTROL,
    ExperimentError,
    Variant,
    VideoCandidates,
    compare,
    compare_unlabeled,
    evaluate_variant,
    load_video_candidates,
    predict,
    record_experiment,
)
from backend.infrastructure.database import Database
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

132. assistant (2026-07-02T06:11:33.313Z)

`Sequence` stays in `ablation.py` (used in 5 other signatures) but must be **removed** from `fusion_compare.py` (only the predicate used it). All references are contained. Let me read the exact `VideoCandidates` block in `experiment.py` to place the canonical predicate.

133. assistant (2026-07-02T06:11:33.313Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\experiment.py
[/external_agent_tool_call]

134. assistant (2026-07-02T06:11:33.313Z)

[external_agent_tool_result]
```
# ----------------------------------------------------------------- persisted candidates


@dataclass
class VideoCandidates:
    """One video's persisted detection, ready for offline re-scoring.

    ``intervals`` are the candidate windows; ``features`` is column-major
    (``feature_name -> per-candidate value``, each list aligned to ``intervals``);
    ``gts`` are golden labels (None for new/unlabeled footage). Build one from the DB
    with `load_video_candidates`, or directly in tests.
    """

    name: str
    intervals: List[Interval]
    features: Dict[str, List[float]] = field(default_factory=dict)
    gts: Optional[List[Interval]] = None


def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
    """Persisted feature column, defaulting missing/short columns to 0.0 (matches
    `training.extract_yolo_features`, which lets a sparse/old row still vectorise)."""
    col = vc.features.get(name)
    n = len(vc.intervals)
    if col is None:
        logger.warning("variant references feature %r absent from %s ? treating as 0.0",
                       name, vc.name)
        return [0.0] * n
    if len(col) != n:
        raise ExperimentError(
            f"feature {name!r} has {len(col)} values but {vc.name} has {n} candidates")
    return [float(x) for x in col]


def _feature_matrix(vc: VideoCandidates, feature_names: Sequence[str]) -> List[List[float]]:
    cols = [_feature_column(vc, name) for name in feature_names]
```
[/external_agent_tool_result]

135. assistant (2026-07-02T06:11:33.313Z)

The placement is clear ? right after the `VideoCandidates` dataclass. Let me make all the edits. First, add the canonical predicate to `experiment.py`:

136. assistant (2026-07-02T06:11:33.313Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\experiment.py
[/external_agent_tool_call]

137. assistant (2026-07-02T06:11:33.313Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\experiment.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

138. assistant (2026-07-02T06:11:33.313Z)

Now `ablation.py` ? import the shared predicate and remove its local copy:

139. assistant (2026-07-02T06:11:33.313Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\ablation.py
[/external_agent_tool_call]
```

140. assistant (2026-07-02T06:11:33.313Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\ablation.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

141. assistant (2026-07-02T06:11:33.313Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\ablation.py
[/external_agent_tool_call]
```

142. assistant (2026-07-02T06:11:33.313Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\ablation.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

143. assistant (2026-07-02T06:11:33.313Z)

Now `fusion_compare.py` ? drop `Sequence`, import the shared predicate, remove the local def.
Let me read its current header:

144. assistant (2026-07-02T06:11:33.313Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_compare.py
[/external_agent_tool_call]
```

145. assistant (2026-07-02T06:11:33.313Z)

```
[external_agent_tool_result]
from __future__ import annotations

import argparse
import glob
import json
import logging
import os
from typing import Any, Dict, List, Sequence

from backend.eval import experiment
from backend.eval.eval_stats import paired_delta_ci
from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES

logger = logging.getLogger(__name__)


def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
    """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign test
    (eval_stats, C1 ? "never a bare mean"). Aligns the two variants' per-video held-out
```

```
scores
        by video name, so the headline ?F1 comes with how-wide + did-it-win-on-enough-videos."""
        by_a = {p["video"]: p[metric] for p in res["variant_a"]["per_video"]}
        by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
        vids = [p["video"] for p in res["variant_a"]["per_video"]]
        return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])


    def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) -> bool:
        """True iff every candidate row carries the per-cue feature dicts the fusion loaders read
        (``shuttle`` + ``person``; ``audio`` stays optional ? it is added later by
        ``fusion_audio.augment`` and older JSONs predate it).

        The velocity-windowing / distill corpus-growth path persists a LEANER
        ``*_golden_features.json`` row (``start``/``end`` + velocity stats, no cue dicts) for clips
        that never went through fusion feature extraction. The loaders SKIP those rather than crash
        on ``c["shuttle"]`` ? so a MIXED corpus (some fusion files, some windowing-only) loads the
        fusion subset cleanly. An empty candidate list is not a fusion file either.

        (Kept local to the fusion loaders; ``ablation.py`` carries its own copy of this one-liner ?
        the two modules are independent and the shared form would force an awkward import.)"""
        return bool(candidates) and all("shuttle" in c and "person" in c for c in candidates)
```
[/external_agent_tool_result]

146. assistant (2026-07-02T06:11:33.313Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_compare.py
[/external_agent_tool_call]

147. assistant (2026-07-02T06:11:33.314Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_compare.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

148. assistant (2026-07-02T06:11:33.314Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_compare.py
[/external_agent_tool_call]

149. assistant (2026-07-02T06:11:33.314Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_compare.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

150. assistant (2026-07-02T06:11:33.314Z)

Now `fusion_audio.py` ? repoint the predicate import from `fusion_compare` to the canonical `experiment`:

151. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_audio.py
[/external_agent_tool_call]
```

152. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_result]

import argparse
import glob
import json
import logging
import os
from typing import Any, Dict, List

from backend.eval import experiment
from backend.eval.fusion_compare import _is_fusion_schema, delta_stats
from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES, _derive_video_path
from backend.pipeline.detectors import audio_features as af
from backend.pipeline.audio.hit_detector import detect_hits_in_video

logger = logging.getLogger(__name__)

[/external_agent_tool_result]
```

153. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_audio.py
[/external_agent_tool_call]
```

154. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\backend\eval\fusion_audio.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

155. assistant (2026-07-02T06:11:33.314Z)

Now the tests. Let me move the canonical predicate unit test to `test_experiment.py` (its true home now). First read `test_experiment.py` structure:

156. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_experiment.py
[/external_agent_tool_call]
```

157. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_result]

from backend.eval.classifier import LogisticRegression
from backend.eval.experiment import (
    CONTROL,
    ExperimentError,
    Variant,
    VideoCandidates,
    compare,
    compare_unlabeled,
    evaluate_variant,
    load_video_candidates,
```

```
    predict,
    record_experiment,
)
from backend.infrastructure.database import Database


# A video whose third candidate is a held-shuttle false fire (net_crossings=0): it does
# not overlap any golden rally, so it is a negative by construction.
def _held_shuttle_video(name="v", gts=None):
    return VideoCandidates(
        name=name,
        intervals=[(0.0, 10.0), (20.0, 30.0), (12.0, 13.0)],
        features={"net_crossings": [4.0, 3.0, 0.0]},
        gts=gts if gts is not None else [(0.0, 10.0), (20.0, 30.0)],
    )


# Three videos where feature "x" perfectly separates rallies (x=5) from junk (x=0),
# so an honest LOVO learned variant should recover F1?1.0 on each held-out video.
def _separable_videos():
    vids = []
    for k in range(3):
        base = 100.0 * k
        vids.append(VideoCandidates(
            name=f"sep{k}",
            intervals=[(base + 0, base + 10), (base + 20, base + 30),
                       (base + 40, base + 41), (base + 50, base + 51)],
            features={"x": [5.0, 5.0, 0.0, 0.0]},
            gts=[(base + 0, base + 10), (base + 20, base + 30)],
        ))
    return vids


# ---------------------------------------------------------------------------- Variant

def test_variant_roundtrip_and_hash_stability():
    v = Variant(name="gateA", min_crossings=2, cue_flags={"person": False})
    assert Variant.from_dict(v.to_dict()) == v
    assert v.spec_hash() == v.spec_hash()                      # stable
    assert CONTROL.spec_hash() == Variant(name="control").spec_hash()
    assert v.spec_hash() != CONTROL.spec_hash()               # different recipe ?
different hash


def test_from_dict_ignores_unknown_keys():
[/external_agent_tool_result]
```

158. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_experiment.py
[/external_agent_tool_call]
```

159. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_experiment.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

160. assistant (2026-07-02T06:11:33.314Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_experiment.py
```

[/external_agent_tool_call]

161. assistant (2026-07-02T06:11:33.314Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_experiment.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

162. assistant (2026-07-02T06:11:33.314Z)

Now remove the now-duplicated unit test from `test_ablation.py`:

163. assistant (2026-07-02T06:11:33.314Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_ablation.py
[/external_agent_tool_call]

164. assistant (2026-07-02T06:11:33.314Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-consolidate-schema\tests\test_ablation.py has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

165. assistant (2026-07-02T06:11:33.315Z)

Now let me verify `test_ablation.py`'s remaining `ablation._is_fusion_schema` reference still resolves (via re-import) and run the targeted test files:

166. assistant (2026-07-02T06:11:33.315Z)

[external_agent_tool_call: Bash]
description: Run the four affected test files
command: cd "../bhi-consolidate-schema" && echo "=== test_ablation imports ablation as module? ===" && grep -n "import ablation\|from backend.eval import" tests/test_ablation.py | head; echo "=== run targeted tests ===" && python -m pytest tests/test_experiment.py tests/test_ablation.py tests/test_fusion_compare.py tests/test_audio_features.py -q 2>&1 | tail -20
[/external_agent_tool_call]

167. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_result]
=== test_ablation imports ablation as module? ===
25:from backend.eval import ablation
=== run targeted tests ===
.........................................ss...................           [100%]
60 passed, 2 skipped in 23.07s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

168. assistant (2026-07-02T06:11:33.315Z)

All 60 pass (2 skipped = golden-litmus skipif). Now the full verification gate ? suite with coverage, ruff, mypy:

169. assistant (2026-07-02T06:11:33.315Z)

[external_agent_tool_call: Bash]
description: Run full suite with coverage gate
command: cd "../bhi-consolidate-schema" && echo "=== FULL SUITE + COVERAGE ===" && python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70 2>&1 | tail

```
-6
[/external_agent_tool_call]
```

170. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_result]
=== FULL SUITE + COVERAGE ===
TOTAL                                               8009   1546    81%
Required test coverage of 70% reached. Total coverage: 80.70%
855 passed, 7 skipped, 7 warnings in 100.00s (0:01:39)
========================================================================
 RESULT: PASS - all tests green
========================================================================
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

171. assistant (2026-07-02T06:11:33.315Z)

Suite green: **855 passed, 7 skipped**, coverage **80.70%**. Now ruff + mypy:

172. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_call: Bash]
description: Run ruff and mypy gates
command: cd "../bhi-consolidate-schema" && echo "=== RUFF ===" && ruff check . 2>&1 | tail -8;
echo "=== MYPY ===" && mypy backend training 2>&1 | tail -8
[/external_agent_tool_call]
```

173. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_result]
=== RUFF ===
All checks passed!
=== MYPY ===
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs  [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 104 source files
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

174. assistant (2026-07-02T06:11:33.315Z)

All gates green. Let me review the full diff and do a final repo-wide sweep for any stale reference to the old import location.

175. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_call: Bash]
description: Sweep for stale references and show diff stat
command: cd "../bhi-consolidate-schema" && echo "=== STALE REFERENCE SWEEP: any import of
_is_fusion_schema from fusion_compare, or defs outside experiment.py? ===" && grep -rn
"fusion_compare import.*_is_fusion_schema\|def _is_fusion_schema" backend/ tests/ scripts/ docs/
2>/dev/null || echo "(none ? good)"; echo; echo "=== all _is_fusion_schema references now ==="
&& grep -rn "_is_fusion_schema" backend/ tests/ 2>/dev/null; echo; echo "=== DIFF STAT ===" &&
git diff --stat
[/external_agent_tool_call]
```

176. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_result]
=== STALE REFERENCE SWEEP: any import of _is_fusion_schema from fusion_compare, or defs outside
experiment.py? ===
backend/eval/experiment.py:180:def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) ->
bool:

=== all _is_fusion_schema references now ===
backend/eval/ablation.py:57:from backend.eval.experiment import Variant, VideoCandidates,
_is_fusion_schema
backend/eval/ablation.py:394:    ablatable subset cleanly (see ``_is_fusion_schema``)."""
backend/eval/ablation.py:400:        if not _is_fusion_schema(cands):
backend/eval/experiment.py:180:def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) ->
bool:
backend/eval/fusion_audio.py:23:from backend.eval.experiment import _is_fusion_schema
backend/eval/fusion_audio.py:61:    crashed on, so a mixed corpus loads the fusion subset
cleanly (see ``_is_fusion_schema``).
backend/eval/fusion_audio.py:69:        if not _is_fusion_schema(cands):
backend/eval/fusion_compare.py:23:from backend.eval.experiment import _is_fusion_schema
backend/eval/fusion_compare.py:46:    fusion subset cleanly (see ``_is_fusion_schema``)."""
backend/eval/fusion_compare.py:52:        if not _is_fusion_schema(cands):
Binary file backend/eval/__pycache__/ablation.cpython-313.pyc matches
Binary file backend/eval/__pycache__/experiment.cpython-313.pyc matches
Binary file backend/eval/__pycache__/fusion_audio.cpython-313.pyc matches
Binary file backend/eval/__pycache__/fusion_compare.cpython-313.pyc matches
tests/test_ablation.py:408:        if ablation._is_fusion_schema(cands):
tests/test_experiment.py:18:    _is_fusion_schema,
tests/test_experiment.py:58:def test_is_fusion_schema_distinguishes_lean_and_full():
tests/test_experiment.py:66:    assert _is_fusion_schema(full) is True
tests/test_experiment.py:67:    assert _is_fusion_schema(no_audio) is True
tests/test_experiment.py:68:    assert _is_fusion_schema(lean) is False
tests/test_experiment.py:69:    assert _is_fusion_schema([]) is False
Binary file tests/__pycache__/test_ablation.cpython-313-pytest-9.0.3.pyc matches
Binary file tests/__pycache__/test_experiment.cpython-313-pytest-9.0.3.pyc matches

=== DIFF STAT ===
 backend/eval/ablation.py       | 14 +-------------
 backend/eval/experiment.py     | 18 ++++++++++++++++++
 backend/eval/fusion_audio.py   |  3 ++-
 backend/eval/fusion_compare.py | 19 ++-----------------
 tests/test_ablation.py         | 12 ------------
 tests/test_experiment.py       | 17 +++++++++++++++++
 6 files changed, 40 insertions(+), 43 deletions(-)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

177. assistant (2026-07-02T06:11:33.315Z)

Clean: single definition in `experiment.py:180`, all three consumers import it, no stale `fusion_compare` imports. The `.pyc` matches are gitignored `__pycache__`. Let me review the full diff content:

178. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_call: Bash]
description: Show full content diff
command: cd "../bhi-consolidate-schema" && git diff -- backend/eval/ablation.py
backend/eval/fusion_audio.py backend/eval/fusion_compare.py backend/eval/experiment.py
tests/test_ablation.py tests/test_experiment.py
[/external_agent_tool_call]
```

179. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_result]
diff --git a/backend/eval/ablation.py b/backend/eval/ablation.py
```

```
index 968eefd..8324eb2 100644
--- a/backend/eval/ablation.py
+++ b/backend/eval/ablation.py
@@ -54,7 +54,7 @@ from dataclasses import dataclass, field
 from typing import Any, Dict, List, Optional, Sequence, Tuple

 from backend.eval import experiment
-from backend.eval.experiment import Variant, VideoCandidates
+from backend.eval.experiment import Variant, VideoCandidates, _is_fusion_schema

 logger = logging.getLogger(__name__)

@@ -383,18 +383,6 @@ def litmus_config() -> AblationConfig:
 # ---------------------------------------------------------------------------- corpus I/O


-def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) -> bool:
-    """True iff every candidate row carries the per-cue feature dicts the ablation scorer reads
-    (``shuttle`` + ``person``; ``audio`` stays optional ? older JSONs predate it).
-
-    The velocity-windowing / distill path persists a LEANER ``*_golden_features.json`` row
-    (``start``/``end`` + velocity stats, no cue dicts) for clips that never went through fusion
-    feature extraction. ``load_videos`` SKIPS those rather than crash on ``c["shuttle"]`` ? so
a
-    mixed corpus (some fusion files, some windowing-only) loads the ablatable subset cleanly.
-    An empty candidate list is not ablatable either."""
-    return bool(candidates) and all("shuttle" in c and "person" in c for c in candidates)
-
-
 def load_videos(features_dir: str) -> List[VideoCandidates]:
     """Load every FULL-FUSION-SCHEMA ``*_golden_features.json`` into a ``VideoCandidates``
     (intervals + shuttle + person + audio columns + golden gts). Audio columns are optional
diff --git a/backend/eval/experiment.py b/backend/eval/experiment.py
index 6b92f3c..67aae66 100644
--- a/backend/eval/experiment.py
+++ b/backend/eval/experiment.py
@@ -177,6 +177,24 @@ class VideoCandidates:
     gts: Optional[List[Interval]] = None


+def _is_fusion_schema(candidates: Sequence[Dict[str, Any]]) -> bool:
+    """True iff every candidate row carries the per-cue feature dicts the golden-feature
loaders
+    read (``shuttle`` + ``person``; ``audio`` stays optional ? it is added later by
+    ``fusion_audio.augment`` and older JSONs predate it).
+
+    The velocity-windowing / distill corpus-growth path persists a LEANER
+    ``*_golden_features.json`` row (``start``/``end`` + velocity stats, no cue dicts) for clips
+    that never went through fusion feature extraction. The loaders that build cue-rich
+    ``VideoCandidates`` from these files ? ``ablation.load_videos``,
``fusion_compare.load_videos``
+    and ``fusion_audio.load_videos_with_audio`` ? SKIP those rather than crash on
``c["shuttle"]``,
+    so a MIXED corpus (some fusion files, some windowing-only) loads the fusion subset cleanly.
+    An empty candidate list is not a fusion file either.
+
+    Single source of truth for the three loaders (consolidated 2026-06-18 after PRs #209/#210
+    each landed an independent copy)."""
+    return bool(candidates) and all("shuttle" in c and "person" in c for c in candidates)
+
+
 def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
     """Persisted feature column, defaulting missing/short columns to 0.0 (matches
     `training.extract_yolo_features`, which lets a sparse/old row still vectorise)."""
diff --git a/backend/eval/fusion_audio.py b/backend/eval/fusion_audio.py
```

```
index d59dd46..295591f 100644
--- a/backend/eval/fusion_audio.py
+++ b/backend/eval/fusion_audio.py
@@ -20,7 +20,8 @@ import os
 from typing import Any, Dict, List

 from backend.eval import experiment
-from backend.eval.fusion_compare import _is_fusion_schema, delta_stats
+from backend.eval.experiment import _is_fusion_schema
+from backend.eval.fusion_compare import delta_stats
 from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES,
_derive_video_path
 from backend.pipeline.detectors import audio_features as af
 from backend.pipeline....
[/external_agent_tool_result]
```

180. assistant (2026-07-02T06:11:33.315Z)

The diff is clean and complete. Let me do one last sanity check ? confirm no `__all__` exports referenced the removed defs (which wouldn't be import-time errors but would be stale):

181. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_call: Bash]
description: Check for __all__ exports in touched modules
command: cd "../bhi-consolidate-schema" && grep -rn "__all__" backend/eval/fusion_compare.py
backend/eval/ablation.py backend/eval/experiment.py backend/eval/fusion_audio.py || echo "(no
__all__ in any touched module ? nothing to update)"
[/external_agent_tool_call]
```

182. assistant (2026-07-02T06:11:33.315Z)

```
[external_agent_tool_result]
(no __all__ in any touched module ? nothing to update)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

183. assistant (2026-07-02T06:11:33.316Z)

Refactor complete and fully verified. Now commit ? re-verifying HEAD and staging explicit paths (shared checkout):

184. assistant (2026-07-02T06:11:33.316Z)

```
[external_agent_tool_call: Bash]
description: Re-verify branch state before commit
command: cd "../bhi-consolidate-schema" && echo "=== branch ===" && git branch --show-current &&
echo "=== HEAD (should be origin/master tip) ===" && git log --oneline -1 && echo "=== commits
vs origin/master (empty pre-commit) ===" && git log --oneline origin/master..HEAD && echo "===
status (only my 6 files) ===" && git status --short
[/external_agent_tool_call]
```

185. assistant (2026-07-02T06:11:33.316Z)

```
[external_agent_tool_result]
=== branch ===
chore/consolidate-fusion-schema-predicate
=== HEAD (should be origin/master tip) ===
f5f339a fix(eval): fusion loaders skip windowing-only files on a mixed golden corpus (#210)
=== commits vs origin/master (empty pre-commit) ===
=== status (only my 6 files) ===
 M backend/eval/ablation.py
 M backend/eval/experiment.py
 M backend/eval/fusion_audio.py
```

M backend/eval/fusion_compare.py
 M tests/test_ablation.py
 M tests/test_experiment.py
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

186. assistant (2026-07-02T06:11:33.316Z)

Clean ? HEAD at `origin/master` tip, no sibling commits, only my 6 files. Staging explicit paths and committing:

187. assistant (2026-07-02T06:11:33.316Z)

[external_agent_tool_call: Bash]
description: Stage explicit paths and commit
command: cd "../bhi-consolidate-schema" && git add backend/eval/ablation.py backend/eval/experiment.py backend/eval/fusion_audio.py backend/eval/fusion_compare.py tests/test_ablation.py tests/test_experiment.py && git commit -q -F - <<'EOF' && git log --oneline -1
refactor(eval): single source of truth for the _is_fusion_schema predicate

PRs #209 (ablation litmus) and #210 (fusion compare/audio loaders) each
landed an independent, byte-identical copy of `_is_fusion_schema` ? the
guard that lets the golden-feature loaders skip leaner velocity-windowing
files on a mixed corpus. Both are now merged, so the deliberate duplication
(taken to avoid stacking #210 on the unmerged #209) can be retired.

Consolidate to one definition in `experiment.py`, next to `VideoCandidates`
(the type these loaders build and the module all three already import ? so
no new import edges and no circular risk: experiment imports none of
ablation / fusion_compare / fusion_audio). `ablation.load_videos`,
`fusion_compare.load_videos` and `fusion_audio.load_videos_with_audio` now
import the shared predicate; their bodies are unchanged. The predicate's
logic is identical ? pure refactor, no behaviour change.

- Drop the now-unused `Sequence` import from `fusion_compare` (it was added
  only for the local def); `ablation` keeps `Sequence` (used elsewhere).
- Move the predicate's canonical unit test to `test_experiment.py` (its new
  home) and remove the duplicate from `test_ablation.py`. `test_ablation`'s
  `_golden_present` keeps using `ablation._is_fusion_schema` ? valid via the
  re-import, consistent with the existing `_derive_video_path` cross-module
  pattern.

Verified locally: `run_all_tests.py --cov ... --cov-fail-under=70` ? 855
passed, 7 skipped, coverage 80.70%; `ruff check .` clean; `mypy backend
training` clean.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>
EOF
[/external_agent_tool_call]

188. assistant (2026-07-02T06:11:33.316Z)

[external_agent_tool_result]
e268972 refactor(eval): single source of truth for the _is_fusion_schema predicate
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

189. assistant (2026-07-02T06:11:33.316Z)

[external_agent_tool_call: Bash]
description: Verify single LF commit and push
command: cd "../bhi-consolidate-schema" && echo "=== only my commit vs origin/master ===" && git log --oneline origin/master..HEAD && echo "=== LF clean? ===" && (git show HEAD | grep -c $'\r'

| grep -q '^0$' && echo "0 CR ? LF clean" || echo "HAS CR") && echo "=== push ===" && git push -u origin chore/consolidate-fusion-schema-predicate 2>&1 | tail -4
[/external_agent_tool_call]

190. assistant (2026-07-02T06:11:33.316Z)

```
[external_agent_tool_result]
=== only my commit vs origin/master ===
e268972 refactor(eval): single source of truth for the _is_fusion_schema predicate
=== LF clean? ===
0 CR ? LF clean
=== push ===
remote:
branch 'chore/consolidate-fusion-schema-predicate' set up to track 'origin/chore/consolidate-fusion-schema-predicate'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      chore/consolidate-fusion-schema-predicate -> chore/consolidate-fusion-schema-predicate
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

191. assistant (2026-07-02T06:11:33.316Z)

[external_agent_tool_call: Bash]
description: Open the consolidation PR
command: cd "../bhi-consolidate-schema" && gh pr create --base master --head chore/consolidate-fusion-schema-predicate --title "refactor(eval): single source of truth for the _is_fusion_schema predicate" --body "$(cat <<'EOF'

Why

Follow-up diligence cleanup. PRs [#209](https://github.com/avidullu/badminton-highlight-indexer/pull/209) (ablation litmus) and [#210](https://github.com/avidullu/badminton-highlight-indexer/pull/210) (fusion compare/audio loaders) each landed an **independent, byte-identical** copy of `_is_fusion_schema` ? the guard that lets the golden-feature loaders skip leaner velocity-windowing files on a mixed corpus. The duplication was deliberate at the time (to avoid stacking #210 on the then-unmerged #209). Both are now merged on `master`, so it can be retired.

What

Consolidate to **one definition** in `backend/eval/experiment.py`, placed next to `VideoCandidates` ? the type these loaders build, and the module all three already import. This means:
- **No new import edges** (`ablation`, `fusion_compare`, `fusion_audio` already import `experiment`).
- **No circular-import risk** ? `experiment` imports none of the three consumers.

The three loaders ? `ablation.load_videos`, `fusion_compare.load_videos`, `fusion_audio.load_videos_with_audio` ? now import the shared predicate; **their bodies are unchanged** and the predicate logic is identical. Pure refactor, **no behaviour change**.

Details:
- Dropped the now-unused `Sequence` import from `fusion_compare` (it was added only for the local def). `ablation` keeps `Sequence` (still used in 5 other signatures).
- Repointed `fusion_audio`'s predicate import from `fusion_compare` ? `experiment` (single source).
- Moved the predicate's **canonical unit test** to `tests/test_experiment.py` (its new home) and removed the duplicate from `tests/test_ablation.py`. `test_ablation`'s `_golden_present` keeps referencing `ablation._is_fusion_schema` ? valid via the re-import, consistent with the existing `_derive_video_path` cross-module pattern.

Verification (local)

- `python run_all_tests.py --cov=backend --cov=training --cov-fail-under=70` ? **855 passed, 7

skipped**, coverage **80.70%** (?70 floor, no regression).
- `ruff check .` ? clean.
- `mypy backend training` ? clean (no issues in 104 files).
- Repo-wide sweep confirms a single `def _is_fusion_schema` (in `experiment.py`) and no stale imports from the old location.

? Generated with [Claude Code](https://claude.com/claude-code)
EOF
)" 2>&1 | tail -3
[/external_agent_tool_call]

192. assistant (2026-07-02T06:11:33.316Z)

[external_agent_tool_result]
https://github.com/avidullu/badminton-highlight-indexer/pull/215
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

193. assistant (2026-07-02T06:11:33.316Z)

Now clean up the worktree:

194. assistant (2026-07-02T06:11:33.316Z)

[external_agent_tool_call: Bash]
description: Remove the consolidation worktree
command: cd "C:/Users/avidu/Projects/badminton-highlight-indexer" && git worktree remove "../bhi-consolidate-schema" 2>&1 && echo "worktree removed"
[/external_agent_tool_call]

195. assistant (2026-07-02T06:11:33.316Z)

[external_agent_tool_result]
worktree removed
[/external_agent_tool_result]

196. assistant (2026-07-02T06:11:33.316Z)

Done ? the diligence cleanup is up as PR [#215](https://github.com/avidullu/badminton-highlight-indexer/pull/215).

What I did

Confirmed both fixes are merged on `master` (ablation = [#209](https://github.com/avidullu/badminton-highlight-indexer/pull/209), fusion loaders = [#210](https://github.com/avidullu/badminton-highlight-indexer/pull/210)), which left **two byte-identical copies** of `_is_fusion_schema`. Consolidated them to a single source of truth:

- **Canonical definition ? [`backend/eval/experiment.py`](backend/eval/experiment.py)**, right next to `VideoCandidates`. I picked `experiment.py` deliberately over the conceptually-tempting `fusion_golden.py`: all three consumers already import `experiment`, and `experiment` imports none of them ? so **zero new import edges and no circular-import risk**. `fusion_golden` would have dragged its heavy `TrackNetRunner`/detector deps into `ablation.py` for a pure dict-key predicate.
- The three loaders (`ablation.load_videos`, `fusion_compare.load_videos`, `fusion_audio.load_videos_with_audio`) now import the shared predicate; **bodies unchanged, logic identical** ? a true no-behaviour-change refactor.
- Removed the stale `Sequence` import from `fusion_compare` (kept in `ablation`, still used in 5 signatures); repointed `fusion_audio`'s import to the new home.
- Moved the predicate's canonical unit test into `test_experiment.py` and removed the duplicate from `test_ablation.py`.

Verification (isolated worktree off `origin/master`)

- Full suite: **855 passed, 7 skipped**, coverage **80.70%** (?70 floor)
- `ruff check .` clean · `mypy backend training` clean
- Repo-wide sweep: exactly one `def _is_fusion_schema` (in `experiment.py`), no stale imports, no `__all__` to update

Kept it to one self-contained refactor PR, branched from `origin/master`, HEAD re-verified before commit, explicit paths staged, worktree removed after push.

One small judgment call worth flagging: I kept the name underscore-prefixed (`_is_fusion_schema`) rather than promoting it to a public `is_fusion_schema`. It's an internal eval helper, and the codebase already imports underscore-prefixed helpers across modules (e.g. `_derive_video_path`), so this matches the existing convention with less churn. Easy to flip if you'd prefer it public.